

Pressure-preserving refinements for simple zeros of the Riemann zeta function

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Abstract

We prove the unconditional lower bound

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{1125000H_{\text{MT}} - 2220}{1120671} = 0.6731175265883904388095857434106058666\dots,$$

where

$$H_{\text{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}.$$

The proof uses the certified position-weighted seven-point inequality

$$\delta = \frac{39}{10000}, \quad p = \frac{1}{10^7}(2714, 3733, 3553, 3553, 3733, 2714), \quad \beta := \sum_{j=1}^6 p_j = \frac{1}{500}.$$

Two refinements are introduced. First, the pressure carried by each relative gap position is retained through the block summation and the shifted-block average. This replaces the coarse global pressure cost $\beta(m-1)$ by the exact mass $\beta(m-6)$. Second, for a limiting block Gram matrix $K_B = (k(y_i - y_j))_{1 \leq i, j \leq m}$, pinching into the two alternating families of adjacent pairs gives

$$\mathcal{D}(K_B) \geq \sum_{j=1}^{m-1} w(g_j).$$

Combined with a scalar kernel inequality at $g_0 = 89/100$ and $t = 33/2$, this yields

$$\mathcal{D}(G_B) + P_B \geq \frac{39}{10000}(m-6) - o(1)$$

for $m = 450$. Shifted-block averaging then gives the stated constant. The additional signed kernel inequality

$$\frac{171389}{1000000} < k(89/100)$$

is proved analytically in Lean from exact integral identities, directed Taylor bounds for sine and cosine, certified bounds for π already available in Mathlib, and exact rational arithmetic. Positivity and monotonicity then give $w(89/100) > 2937/100000$. The accompanying exact-arithmetic verifier provides an independent reproducibility check of the scalar inequality, the pressure mass, the contradiction margin, and the final constant. The companion Lean development proves the signed scalar kernel inequality analytically and closes the six-gap hard core by an exact rational finite certificate. The finite closure reduces 511 bootstrap codes to 231, then 23, then 5 survivors and rules out the final five. These formal components yield the seven-point inequality used in the paper, the published epsilon-form estimate, and the literal liminf statement above. Eventual positivity of the denominator follows from the formalized Riemann–von Mangoldt theorem. The Arb/FLINT computation is retained only as an independent reproducibility check and is not a premise of the Lean theorem path. The written mathematical derivation below does not invoke the recent Alpöge–Furman theorem as a premise; the software provenance and exact formal trust boundary are stated separately in Section 9.

1 Introduction and statement of novelty

Let $N(T, 2T)$ denote the number of nontrivial zeros of $\zeta(s)$ with ordinary ordinate in $(T, 2T]$, counted with multiplicity, and let $N_0^s(T, 2T)$ denote the number of simple zeros on the critical line in the same interval. The quantitative study of N_0^s/N goes back to Montgomery’s pair-correlation method [17, 18] and the Montgomery–Taylor optimization. Recent work has made the relevant finite-compression and zero-side inertia mechanism unconditional. Alpöge and Furman [5], communicating a proof discovered autonomously by Claude at Anthropic as recorded both by the authors and by Anthropic [6], establish the finite-compression, rank–trace, and inertia route unconditionally. The Montgomery–Taylor clause of their Theorem A gives the optimized simple-zero constant denoted below by H_{MT} . The companion Anthropic Lean artifact formalizes the corresponding Montgomery–Taylor constant and theorem family [9]. Lamzouri [7] subsequently gave a conceptually different unconditional proof of the same 67.25%-level conclusion. For surrounding pair-correlation work see Baluyot–Goldston–Suriajaya–Turnage–Butterbaugh [12, 13] and Goldston–Suriajaya [14, 15]. For the optimized Montgomery–Taylor test family we write

$$H_{\text{MT}} := \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 0.6725007036794116457\dots \quad (1)$$

Two very recent public computational manuscripts are especially close to the present argument. Jo’s reproducible draft [19] obtains

$$0.6730085279277797613\dots$$

using a stability-enhanced rank–trace inequality, a seven-point local functional, and a uniform pressure on its six gaps. A subsequent repository maintained by Rademacher [20], explicitly based on Jo’s public artifact, strengthens the uniform seven-point certificate and obtains

$$0.6730213619501665335\dots$$

Jo’s repository identifies its research draft as generated by GPT-5.6 Sol. Rademacher’s repository records Jo’s artifact as its upstream provenance and presents the subsequent refinement as AI-generated. We cite the human-maintained artifacts and treat model use as provenance rather than mathematical authorship.

As of 21 September 2026, several contemporaneous public computational artifacts report numerically larger values in closely related settings. Ojha’s research repository [10] reports the dyadic candidate

$$0.6733127422722459\dots,$$

and explicitly labels the theorem claim a record candidate pending expert review and end-to-end formalization. Devine’s Zenodo preprint [11] reports the dyadic value

$$0.673399.$$

Gebendorfer’s research draft [21] reports $0.6734760237\dots$ using an expanded analytic window and joint position-pressure vectors; a companion draft [22] reports $0.6734765075\dots$ after a compatible-triple refinement. These drafts use the full-height ratio $N_{\text{sc}}(T)/N(T)$ rather than the dyadic ratio of Theorem 1.1, and identify themselves as internally checked research drafts pending independent review. We therefore cite them as methodologically relevant contemporary work rather than treat the numerical values as directly interchangeable. None of these artifacts is an input to the present

proof, and we make no claim that the numerical value of Theorem 1.1 is the largest publicly reported bound.

The contribution of the present paper is explicit and layered. We do not claim the finite-compression/rank–trace mechanism, the stability-enhanced rank–trace inequality, or the seven-point architecture as new. The position-weighted seven-point inequality used throughout the paper is documented in the reproducibility artifact [23]. Its nonuniform pressure vector is

$$p = \frac{1}{10^7}(2714, 3733, 3553, 3553, 3733, 2714), \quad \beta := \sum_{j=1}^6 p_j = \frac{1}{500}, \quad (2)$$

for which the seven-point functional satisfies

$$\mathcal{F}_p(g_1, \dots, g_6) \geq \frac{39}{10000} \quad (g_j \geq 0). \quad (3)$$

The new argument has two stages. First, rather than replacing the accumulated position-dependent pressure inside a block by $\beta \text{span}(B)$, we retain its exact coefficients until the final average over shifted blocks. This reduces the global pressure mass from the coarse $\beta(m-1)$ to the exact $\beta(m-6)$. Already at $m = 262$ this gives

$$\frac{655000H_{\text{MT}} - 1280}{652504} = 0.6731115225500757511923586 \dots$$

Second, an adjacent-pair pinching argument supplies an independent lower bound for the Gram defect in terms of the consecutive-gap kernel energies. Together with a scalar Montgomery–Taylor kernel inequality proved analytically in Lean, this permits $m = 450$ and yields Theorem 1.1. Thus the numerical improvement in the main theorem comes from the global pressure accounting and the additional Gram-defect information, with (3) as the fixed local inequality.

Theorem 1.1 (Main theorem). *With*

$$H_{\text{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}},$$

one has unconditionally

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{1125000H_{\text{MT}} - 2220}{1120671} = 0.6731175265883904388095857434106058666 \dots$$

A second contribution is expository and logical rather than a separate priority claim: we reconstruct the analytic/Gabor package needed by the proof from classical theorems and explicitly identify the common finite compression on its zero and prime sides. Consequently, Theorem 1.1 does not invoke the Montgomery–Taylor clause of Theorem A of Alpöge–Furman [5], or any other theorem of that paper, as a mathematical premise. This is a statement about the logical derivation presented in the manuscript, not a claim of clean-room software independence: the Lean verification has the distinct provenance recorded in Section 9.

Priority and provenance convention. The proof discovered by Claude and verified and communicated by Alpöge–Furman supplies the immediate unconditional finite-compression/rank–trace comparison point [5, 6]. Jo’s repository is the public upstream artifact for the stability-enhanced seven-point architecture used here, and Rademacher’s repository records the subsequent uniform strengthening. The present paper uses the position-weighted inequality (2)–(3) as its local ingredient;

Proposition 5.2 and the companion formal certificate supply its proof. The global contribution is the exact pressure-preserving accounting, the adjacent-pair defect lemma, the analytic scalar-kernel bound, and the resulting constant, together with a kernel-checked exact closure of the finite six-gap problem. We make no claim that position-dependent pressure, considered as a general strategy, originates here.

Component	Provenance used in this paper
Classical analytic inputs	Weil explicit formula, Montgomery–Taylor optimization, Riemann–von Mangoldt, PNT, Poisson summation, Montgomery–Vaughan and standard matrix analysis
Finite compression, zero-side inertia and rank–trace route	Proof discovered by Claude (Anthropic), verified and communicated by Alpöge–Furman [5, 6]; not imported as a premise of Theorem 1.1
Stability-enhanced rank–trace and seven-point architecture	Jo [19]
Uniform seven-point strengthening	Rademacher [20]
Position-weighted seven-point inequality 39/10000	Hurtado; reproducibility artifact [23], with Arb/FLINT retained as an independent interval check
Exact pressure preservation, adjacent-pair pinching, scalar kernel argument and $m = 450$ assembly	Present paper
Lean base library	Anthropic <code>formal-math/zeta23</code> , pinned commit recorded in Section 9 [9]
Lean signed-kernel theorem	Exact integral identities, directed trigonometric bounds, certified bounds for π , and rational arithmetic
Lean exact seven-point closure	Exact rational E3 reduction $511 \rightarrow 231 \rightarrow 23 \rightarrow 5 \rightarrow \emptyset$, yielding the article seven-point inequality
Lean published endpoint	The companion development proves the epsilon form and the literal liminf statement; formalized Riemann–von Mangoldt gives eventual positivity of $N(T, 2T)$

2 The common finite compression and its two representations

Put

$$\ell = \log \frac{T}{2\pi}, \quad L = \ell, \quad h = \frac{2\pi}{L}, \quad d = \left\lfloor \frac{T}{h} \right\rfloor, \quad \tau_j = T + jh \quad (0 \leq j < d).$$

Let $\phi = \phi_L \in C_c^2(\mathbb{R})$ be the real, even Montgomery–Taylor window supported on $[-L/2, L/2]$, and define

$$\widehat{\phi}(z) := \int_{\mathbb{R}} \phi(u) e^{-izu} du, \quad \mathcal{F}_+ f(z) := \int_{\mathbb{R}} f(u) e^{izu} du.$$

Set

$$f_j(u) := \phi(u) e^{-i\tau_j u}. \tag{4}$$

Since ϕ is even, $\widehat{\phi}$ is even entire, hence

$$F_j(z) := \mathcal{F}_+ f_j(z) = \widehat{\phi}(\tau_j - z) = \widehat{\phi}(z - \tau_j). \tag{5}$$

For a zero $\rho = \beta + i\gamma$ put

$$z_\rho := -i \left(\rho - \frac{1}{2} \right) = \gamma - i \left(\beta - \frac{1}{2} \right). \tag{6}$$

For the horizontal partner $\rho^\# := 1 - \bar{\rho}$,

$$z_{\rho^\#} = \overline{z_\rho}. \quad (7)$$

Let W denote the Hermitian Weil form in the normalization of Appendix C. Define the single common finite compression

$$G_{jk} := W(f_j, f_k), \quad 0 \leq j, k < d. \quad (8)$$

For $c = (c_0, \dots, c_{d-1})^t$, put

$$f_c := \sum_{j=0}^{d-1} \overline{c_j} f_j, \quad \Lambda_\rho(c) := \mathcal{F}_+ f_c(z_\rho) = \sum_{j=0}^{d-1} \overline{c_j} F_j(z_\rho).$$

With this convention, which is chosen only to match the standard matrix quadratic form, one has $c^* G c = W(f_c, f_c)$ exactly.

Proposition 2.1 (Two representations of the common compression). *Let $X = e^L = T/(2\pi)$ and let $\nu_X = \mu + P_X + \Pi_X$ be the density derived from the classical Weil explicit formula in Proposition C.1. Then:*

(a) Spectral representation.

$$G_{jk} = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_j) \overline{\widehat{\phi}(\tau - \tau_k)} \nu_X(\tau) d\tau. \quad (9)$$

(b) Zero-side representation. For every $c \in \mathbb{C}^d$,

$$c^* G c = \sum_{\rho} m_{\rho} \Lambda_{\rho}(c) \overline{\Lambda_{\rho^\#}(c)}. \quad (10)$$

Grouping under $\rho \leftrightarrow \rho^\#$ gives

$$\begin{aligned} c^* G c &= \sum_{\rho=\rho^\#} m_{\rho} |\Lambda_{\rho}(c)|^2 \\ &\quad + \sum_{\{\rho, \rho^\#\}, \rho \neq \rho^\#} 2m_{\rho} \Re\left(\Lambda_{\rho}(c) \overline{\Lambda_{\rho^\#}(c)}\right). \end{aligned} \quad (11)$$

(c) Interior/exterior decomposition. Let $I' = (T - T^{1/2}, 2T + T^{1/2}]$. Define A by restricting the zero-side sum to zeros with ordinary ordinates in I' , and E by the complementary sum. Since $\rho \mapsto \rho^\#$ preserves ordinary ordinate,

$$G = A + E \quad (12)$$

entrywise.

(d) Normalized evaluation vectors. Put

$$a = \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du, \quad \widehat{A} = \frac{A}{aL^2}.$$

For a simple critical-line zero ρ in I' , define

$$v_{\rho} := \frac{1}{\sqrt{a}L} (F_0(\gamma), \dots, F_{d-1}(\gamma))^t.$$

Then $\Lambda_\rho(c)/(\sqrt{a}L) = c^*v_\rho$, and

$$\|v_\rho\|^2 = \frac{1}{aL^2} \sum_{j=0}^{d-1} |\hat{\phi}(\gamma - \tau_j)|^2 \leq 1. \quad (13)$$

Thus

$$P_1 := \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^*$$

is exactly the normalized simple-critical-line part of $\hat{A}$, and $Q' := \hat{A} - P_1$ is the pull-back of the remaining zero-side blocks.

Proof. Proposition C.1 and (5) give (9). The zero-side form of the same Weil identity is

$$W(f, g) = \sum_{\rho} m_{\rho} \mathcal{F}_+ f(z_{\rho}) \overline{\mathcal{F}_+ g(\overline{z_{\rho}})}.$$

By the coefficient convention above, $c^*Gc = W(f_c, f_c)$. Using (7) with $f = g = f_c$ therefore proves (10); grouping paired zeros proves (11). Splitting this same zero sum by whether the ordinary ordinate lies in I' proves (12).

For a critical-line zero, $z_{\rho} = \gamma \in \mathbb{R}$. Extending the finite sum in (13) to the full translated critical lattice and applying Lemma D.2 at equal arguments (translation of the lattice changes only nonzero dual phases, whose alias terms vanish by exact support) gives

$$\sum_{j \in \mathbb{Z}} |\hat{\phi}(\gamma - \tau_j)|^2 = aL^2.$$

Hence the finite sum is at most aL^2 . Finally, (11) identifies the simple part with $\sum v_{\rho} v_{\rho}^*$ and the remainder with the pull-back of the multiple and off-line blocks. $\square$

Remark 2.2 (Dependency graph). *Input I uses the zero-side representation above, Input II uses $G = A + E$, Input III uses the spectral representation, and Input IV supplies the critical-lattice identity used in (13). Thus all four modules refer to one test family and one matrix.*

3 Analytic/Gabor package and the stability refinement

Put

$$\ell := \log \frac{T}{2\pi}, \quad I = (T, 2T], \quad D_0 = T^{1/2}, \quad I' = (T - D_0, 2T + D_0],$$

and set $L = \ell$. Let ϕ be the real, even, compactly tapered Montgomery–Taylor window used below, and define

$$a := \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du, \quad \hat{G} := \frac{G}{aL^2}, \quad \hat{A} := \frac{A}{aL^2}.$$

The analytic work needed by the stability argument is summarized in the next proposition. Its four assertions are derived in Appendices A–D from the trust base stated below.

Proposition 3.1 (Analytic/Gabor package). *Let $N(I')$ denote the number of nontrivial zeros with ordinates in I' , counted with analytic multiplicity. Partition the distinct zeros in I' into simple critical-line zeros $\mathcal{S}_1$, distinct multiple critical-line zeros $\mathcal{S}_2$, and unordered horizontal pairs*

$$\{\rho, \rho^{\#}\}, \quad \rho^{\#} = 1 - \bar{\rho},$$

off the critical line. Write $s_1 = \#\mathcal{S}_1$, $s_2 = \#\mathcal{S}_2$ and $p = \#\mathcal{P}$. Then the following hold.

(I) Zero-side inertia.

$$N(I') \geq s_1 + 2s_2 + 2p. \quad (14)$$

For the normalized sampled vectors v_ρ associated with $\rho \in \mathcal{S}_1$, let V have these vectors as columns and put

$$P_1 = VV^*, \quad M = V^*V, \quad Q' = \hat{A} - P_1.$$

Then

$$n_+(Q') \leq s_2 + p. \quad (15)$$

(II) Exterior-zero tail. Writing $\hat{A} = \hat{G} - \hat{E}$,

$$4 \operatorname{tr} \hat{A} - \|\hat{A}\|_{\mathbb{F}}^2 \geq 4 \operatorname{tr} \hat{G} - \|\hat{G}\|_{\mathbb{F}}^2 - o(N(T, 2T)). \quad (16)$$

In fact the proof gives $\|\hat{E}\|_1 \ll T^{-1/2}$.

(III) Optimized first and second moments.

$$\operatorname{tr} \hat{G} = N(T, 2T)(1 + o(1)), \quad \|\hat{G}\|_{\mathbb{F}}^2 = \left(\frac{1}{c_1^*} + o(1) \right) N(T, 2T), \quad (17)$$

where

$$\frac{1}{c_1^*} = \frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}, \quad 2 - \frac{1}{c_1^*} = H_{\text{MT}}. \quad (18)$$

(IV) Critical-lattice Poisson–Gabor limit. At mesh $h = 2\pi/L$, Poisson summation gives the infinite-grid identity without nonzero aliases. The tapered Fourier decay controls finite-grid truncation, and the normalized overlap kernel converges to the compact Montgomery–Taylor kernel used in Section 4. The integrated second-moment end effect is

$$O\left(L\ell \log L (\ell^2 + X)\right), \quad X = e^L \asymp T,$$

which is $o(TL\ell^2)$ at $L = \ell$.

Proof. Assertion (I) is Theorem A.5; assertion (II) is Theorem B.6; assertion (III) is proved in Appendix C, culminating in (95); and assertion (IV) is Theorem D.1. These appendices use only the classical trust base listed in Section 9. $\square$

The stability-enhanced rank–trace inequality used next is due to Jo [19]; we include the proof to fix the normalization and the precise defect functional used in the present argument.

Define

$$\Psi(t) = \begin{cases} (t-1)^2, & 0 \leq t \leq 2, \\ 2t-3, & t \geq 2, \end{cases} \quad \mathcal{D}(M) := \operatorname{tr} \Psi(M) \quad (19)$$

for $M \succeq 0$. The function Ψ is nonnegative and convex on $[0, \infty)$: its derivative on $[0, 2]$ is $2(t-1)$, the affine branch has slope 2, and the one-sided derivatives agree at $t = 2$. This is the only convexity property required by the pinching arguments below.

Lemma 3.2 (Stability-enhanced rank–trace inequality). *Let V be a $d \times r$ matrix whose columns have norm at most one. Put $P = VV^* \succeq 0$ and $M = V^*V$. If Q is Hermitian and $n_+(Q) \leq b$, then*

$$\|P + Q\|_{\mathbb{F}}^2 \geq 4 \operatorname{tr}(P + Q) - 3r - 4b + \mathcal{D}(M). \quad (20)$$

Proof. Write $Q = Q_+ - Q_-$ for its positive and negative parts. They are positive semidefinite, have orthogonal supports, and $\text{rank } Q_+ \leq b$. Since $\text{tr}((P - Q_-)Q_+) = \text{tr}(PQ_+) \geq 0$,

$$\|P + Q\|_F^2 \geq \|P - Q_-\|_F^2 + \|Q_+\|_F^2.$$

If $q > 0$, then

$$q^2 - (4q - 4) = (q - 2)^2 \geq 0.$$

If Q_+ has $k \leq b$ positive eigenvalues, summing this inequality gives

$$\|Q_+\|_F^2 \geq 4 \text{tr } Q_+ - 4k \geq 4 \text{tr } Q_+ - 4b.$$

Thus

$$\|Q_+\|_F^2 \geq 4 \text{tr } Q_+ - 4b. \quad (21)$$

Let $p_1 \geq \dots \geq p_r \geq 0$ be the eigenvalues of $M = V^*V$ (counted with multiplicity, hence including any zeros). The nonzero eigenvalues of $P = VV^*$ are the same. Pad the spectra of P and Q_- by zeros to a common length $q \geq \max\{d, r\}$, keeping both lists in nonincreasing order; thus $p_i = 0$ for $i > r$. Von Neumann's trace inequality [8] gives

$$\text{tr}(PQ_-) \leq \sum_{i=1}^q p_i n_i = \sum_{i=1}^r p_i n_i.$$

Using the spectral formulas for the Frobenius norm and trace therefore gives

$$\|P - Q_-\|_F^2 + 4 \text{tr } Q_- \geq \sum_{i=1}^q ((p_i - n_i)^2 + 4n_i).$$

For $i > r$ one has $p_i = 0$, so the corresponding term is $n_i^2 + 4n_i \geq 0$. Dropping those terms yields

$$\|P - Q_-\|_F^2 + 4 \text{tr } Q_- \geq \sum_{i=1}^r ((p_i - n_i)^2 + 4n_i).$$

For fixed $p \geq 0$,

$$\min_{n \geq 0} ((p - n)^2 + 4n) = \begin{cases} p^2, & 0 \leq p \leq 2, \\ 4p - 4, & p \geq 2, \end{cases} = 2p - 1 + \Psi(p).$$

Therefore

$$\|P - Q_-\|_F^2 \geq 2 \text{tr } P - r + \mathcal{D}(M) - 4 \text{tr } Q_-. \quad (22)$$

Combining (21) and (22),

$$\|P + Q\|_F^2 \geq 4 \text{tr}(P + Q) - 2 \text{tr } P - r - 4b + \mathcal{D}(M).$$

Finally $\text{tr } P$ is the sum of the squared column norms of V , so $\text{tr } P \leq r$, proving (20). $\square$

Before stating the global consequence, define the retained central set

$$\mathcal{S}^\circ := \{\rho \in \mathcal{S}_1 : T < \gamma_\rho \leq 2T \text{ and } \rho \text{ lies outside the two boundary strips of Section 4}\}. \quad (23)$$

Let $\mathcal{S}^\circ := \#\mathcal{S}^\circ$ and let

$$M^\circ := (\langle v_\rho, v_{\rho'} \rangle)_{\rho, \rho' \in \mathcal{S}^\circ}. \quad (24)$$

Thus M° is a principal submatrix of M and contains only simple critical-line zeros whose ordinates lie in $(T, 2T]$ and for which the compact kernel approximation is uniform.

Corollary 3.3 (Global Gram-defect inequality). *Then*

$$N_0^s(T, 2T) \geq H_{\text{MT}} N(T, 2T) + \mathcal{D}(M^\circ) - o(N(T, 2T)). \quad (25)$$

Proof. Apply Lemma 3.2 to $\hat{A} = P_1 + Q'$ with $r = s_1$ and $b = s_2 + p$. By (14),

$$2N(I') \geq 2s_1 + 4s_2 + 4p,$$

hence

$$3s_1 + 4s_2 + 4p \leq s_1 + 2N(I').$$

Together with (15), this gives

$$\begin{aligned} \|\hat{A}\|_{\text{F}}^2 &\geq 4 \operatorname{tr} \hat{A} - 3s_1 - 4s_2 - 4p + \mathcal{D}(M) \\ &\geq 4 \operatorname{tr} \hat{A} - s_1 - 2N(I') + \mathcal{D}(M). \end{aligned}$$

Hence

$$s_1 \geq 4 \operatorname{tr} \hat{A} - \|\hat{A}\|_{\text{F}}^2 - 2N(I') + \mathcal{D}(M). \quad (26)$$

Apply the independently proved tail estimate (16). Pinch M to the retained central principal block and its complementary principal block. By the standard pinching/majorization principle for Hermitian matrices [8], pinching is a random-unitary average and the spectral trace functional $X \mapsto \operatorname{tr} \Psi(X)$ is convex and unitarily invariant; no operator convexity of Ψ is required. Therefore

$$\mathcal{D}(M) \geq \mathcal{D}(M^\circ) + \mathcal{D}(M^{\text{rest}}) \geq \mathcal{D}(M^\circ),$$

because $M \succeq 0$ and $\Psi(t) \geq 0$ for $t \geq 0$. Moreover,

$$N(I') = N(T, 2T) + o(N(T, 2T)).$$

Because s_1 counts simple critical-line zeros in the enlarged interval I' , whereas $N_0^s(T, 2T)$ counts only those in I , we also use

$$s_1 \leq N_0^s(T, 2T) + N(I' \setminus I). \quad (27)$$

The standard local zero-count estimate gives

$$N(I' \setminus I) \ll D_0 \log T = T^{1/2} \log T = o(N(T, 2T)),$$

and hence

$$N_0^s(T, 2T) \geq s_1 - o(N(T, 2T)). \quad (28)$$

Substitution of (16) and (17) into (26) gives

$$\begin{aligned} s_1 &\geq \left(4 - \frac{1}{c_1^*}\right) N(T, 2T) - 2N(I') + \mathcal{D}(M^\circ) - o(N) \\ &= \left(2 - \frac{1}{c_1^*}\right) N(T, 2T) + \mathcal{D}(M^\circ) - o(N), \end{aligned}$$

because $N(I') = N(T, 2T) + o(N)$. By (18), $2 - 1/c_1^* = H_{\text{MT}}$. Combining this with (28) proves (25). $\square$

4 The limiting overlap kernel

Put

$$h := \frac{2\pi}{L}, \quad x_\rho := \frac{\gamma_\rho - T}{h} = \frac{L(\gamma_\rho - T)}{2\pi}.$$

Define

$$K(x) := \int_{-1/2}^{1/2} \cos(\sqrt{2}t) \cos(2\pi xt) dt \quad (29)$$

$$= \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}}, \quad (30)$$

$$k(x) := \frac{K(x)}{K(0)}, \quad w(x) := k(x)^2, \quad K(0) = \sqrt{2} \sin(1/\sqrt{2}). \quad (31)$$

The apparent singularities in (30) are removable.

Lemma 4.1 (Compact overlap limit). *Fix $R_0 > 0$. Delete simple zeros lying in boundary strips of normalized width L^2 at the two ends of the sampling grid. The number deleted is $o(N(T, 2T))$. Uniformly for retained simple zeros satisfying $|x_\rho - x_{\rho'}| \leq R_0$,*

$$\langle v_\rho, v_{\rho'} \rangle = k(x_\rho - x_{\rho'}) + o(1). \quad (32)$$

Proof. By Appendix D, the independently proved Poisson–Gabor sampling identity, applied to the full grid, gives

$$\frac{1}{aL^2} \sum_{j \in \mathbb{Z}} \widehat{\phi}(\gamma - \tau_j) \widehat{\phi}(\gamma' - \tau_j) = \frac{\Phi(\gamma - \gamma')}{aL}, \quad \Phi = \widehat{\phi^2}.$$

We record explicitly the pointwise truncation estimate needed here. The Fourier decay of the fixed-width tapered window gives, uniformly in L ,

$$|\widehat{\phi}(r)| \ll \min \left\{ L, \frac{1}{|r|}, \frac{1}{r^2} \right\}.$$

Suppose that both points lie at normalized distance at least L^2 from the finite grid ends and that $|x_\rho - x_{\rho'}| \leq R_0$. If an omitted grid index lies $n \geq L^2$ grid steps beyond the nearer end, then

$$|\gamma - \tau_j| \asymp \frac{n}{L}, \quad |\gamma' - \tau_j| \asymp \frac{n}{L}$$

uniformly for fixed R_0 . Hence

$$|\widehat{\phi}(\gamma - \tau_j) \widehat{\phi}(\gamma' - \tau_j)| \ll \frac{L^4}{n^4}.$$

Summing the two omitted tails gives

$$\sum_{n \geq L^2} \frac{L^4}{n^4} \ll L^{-2}.$$

For the optimized window, dominated convergence also gives

$$a = \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du = K(0) + o(1), \quad K(0) = \sqrt{2} \sin(1/\sqrt{2}) > 0,$$

so in particular $a \asymp 1$. After multiplication by the Gram normalization $1/(aL^2)$, the omitted tails therefore contribute $O(L^{-4}) = o(1)$, uniformly for bounded normalized separation. Thus the full-grid Poisson–Gabor identity may be replaced by the finite sampled inner product with a uniform $o(1)$ error.

For $\gamma - \gamma' = hx$ and fixed x , substitute $u = Lt$:

$$\frac{\Phi(hx)}{aL} = \frac{\int_{-1/2}^{1/2} \phi(Lt)^2 e^{-2\pi i x t} dt}{\int_{-1/2}^{1/2} \phi(Lt)^2 dt}.$$

The squared tapered window converges in L^1 to $\cos(\sqrt{2}t)\mathbf{1}_{[-1/2, 1/2]}(t)$; the rescaled taper occupies length $O(L^{-1})$. Hence the last quotient converges to $K(x)/K(0)$, uniformly for x in compact sets. Combining this with the finite-grid tail bound proves (32).

Each deleted boundary strip has ordinate length $O(L)$, and the standard local zero-count estimate $N(t+1) - N(t) \ll \log(t+3)$ shows that only $O(L^2) = o(N)$ zeros are removed. $\square$

Lemma 4.2 (Block defect). *If $G \succeq 0$ is Hermitian, then*

$$\mathrm{tr} \Psi(G) \geq \min \left\{ 1, 2 \sum_{i < j} |G_{ij}|^2 \right\}. \quad (33)$$

Proof. If all eigenvalues of G lie in $[0, 2]$, then $\Psi(G) = (G - I)^2$ and therefore

$$\mathrm{tr} \Psi(G) = \|G - I\|_{\mathrm{F}}^2 = \sum_i (G_{ii} - 1)^2 + 2 \sum_{i < j} |G_{ij}|^2 \geq 2 \sum_{i < j} |G_{ij}|^2.$$

If some eigenvalue $\lambda > 2$, then its contribution is $\Psi(\lambda) = 2\lambda - 3 > 1$, while all other contributions are nonnegative. This proves (33). $\square$

5 Position-weighted seven-point stability

Let $g_1, \dots, g_6 \geq 0$ be six consecutive gaps. For $1 \leq r \leq 6$ and $1 \leq i \leq 7 - r$, write

$$s_{i,r} = g_i + \dots + g_{i+r-1}$$

and define

$$\mathcal{E}_7(g) := \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(s_{i,r}). \quad (34)$$

For $p_j \geq 0$ put

$$\mathcal{F}_p(g) := \sum_{j=1}^6 p_j g_j + \mathcal{E}_7(g). \quad (35)$$

Proposition 5.1 (Exact pressure redistribution). *Assume*

$$\mathcal{F}_p(g) \geq \delta \quad (g \in \mathbb{R}_{\geq 0}^6), \quad \beta := \sum_{j=1}^6 p_j.$$

Let $y_1 < \dots < y_m$, $m \geq 7$, and put

$$g_r := y_{r+1} - y_r \quad (1 \leq r \leq m-1).$$

Define

$$q_r := \sum_{a=1}^{m-6} \sum_{j=1}^6 p_j \mathbf{1}_{\{r=a+j-1\}}, \quad P_B := \sum_{r=1}^{m-1} q_r g_r. \quad (36)$$

Then

$$E_m + P_B \geq \delta(m-6), \quad E_m := 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i). \quad (37)$$

Moreover,

$$\boxed{\sum_{r=1}^{m-1} q_r = \beta(m-6)}, \quad \min_j p_j \leq q_r \leq \beta. \quad (38)$$

Proof. Sum the local inequality over the $m-6$ consecutive seven-point windows. A fixed pair separated by $s \leq 6$ consecutive gaps appears in at most $7-s$ such windows and has coefficient $2/(7-s)$ in each appearance. Its accumulated coefficient is therefore at most 2, which is the coefficient available in E_m . The pressure terms add exactly to P_B , proving (37).

For the mass identity, sum first over windows:

$$\sum_{r=1}^{m-1} q_r = \sum_{a=1}^{m-6} \sum_{j=1}^6 p_j = \beta(m-6).$$

For a fixed global gap, the relative position j can occur at most once, so $q_r \leq \sum_j p_j = \beta$. Every global gap belongs to at least one of the consecutive seven-point windows, hence $q_r \geq \min_j p_j$. $\square$

Proposition 5.2 (Seven-point inequality and certification). *For the pressure vector (2), every $g_1, \dots, g_6 \geq 0$ satisfies*

$$\begin{aligned} & \frac{2714(g_1 + g_6) + 3733(g_2 + g_5) + 3553(g_3 + g_4)}{10^7} \\ & + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}) \geq \frac{39}{10000}. \end{aligned} \quad (39)$$

Exact formal certificate and independent interval check. The companion Lean certificate proves the same inequality by combining the analytic signed-kernel theorem with an exact rational closure of the remaining six-gap hard core. The reduction to that hard core is explicit. Since every pressure coefficient is at least $2714/10^7$, a total span of at least 14.37 is already settled by the pressure term, with exact margin

$$\frac{2714}{10^7} \frac{1437}{100} - \frac{39}{10000} = \frac{9}{500000000} > 0.$$

On the other hand, if any gap satisfies $0 \leq g_j \leq 0.89$, the analytic signed-kernel bound and monotonicity give

$$w(g_j) > \frac{2937}{100000},$$

so its adjacent-pair contribution alone is $\frac{1}{3}w(g_j) > 39/10000$. Thus a strict counterexample would have to satisfy

$$g_j > 0.89 \quad (1 \leq j \leq 6), \quad g_1 + \dots + g_6 < 14.37.$$

The one-body localization and rational micro-floor inequalities then place such a counterexample in one of 511 exact bootstrap cells. Exact discard/contraction certificates give

$$511 \longrightarrow 231 \longrightarrow 23 \longrightarrow 5,$$

and exact rational lower bounds close the final five survivors. The composed Lean theorem is exactly the seven-point inequality stated in Proposition 5.2; no Arb/FLINT enclosure is used in that theorem path.

As an independent certification route, the Arb/FLINT verifier performs exhaustive interval branch-and-bound through `python-flint`, using Arb midpoint-radius interval arithmetic [16]. The pressure coefficients, target, and grid endpoints are exact rationals. The verifier reconstructs rigorous Arb enclosures for w and its required derivatives from (30)–(31). The grid mesh is $1/4000$. Boxes are discarded only after direct interval evaluation, the positive pressure bound, or a certified tangent bound proves the target. Otherwise the widest coordinate is bisected. An uncertified terminal grid cell raises an exception, so a successful run closes the whole finite search.

The unbounded orthant is reduced by the exact pressure cutoff

$$\frac{2714\,57480}{10^7\,4000} = \frac{1950009}{500000000} = \frac{39}{10000} + \frac{9}{500000000} > \frac{39}{10000}.$$

Thus any box with at least one coordinate beyond the cutoff is certified by the pressure term alone. The independent verifier checks this rational identity before beginning the finite search.

The 256-bit run reports `verified=true`, with 1,119,372 visited nodes, 559,038 splits, and maximum depth 39. The same verifier also succeeds at 128-bit working precision. The complete 256-bit artifact is stored in the companion repository snapshot [23], including the verifier, execution log, environment lock, and cryptographic hashes. Appendix E records the corresponding metadata. $\square$

Corollary 5.3 (Coarse span consequence). *For every $m \geq 7$ and $y_1 < \dots < y_m$,*

$$E_m + \beta(y_m - y_1) \geq \delta(m - 6).$$

For the certified vector (2),

$$E_m + \frac{1}{500}(y_m - y_1) \geq \frac{39}{10000}(m - 6).$$

Proof. Since $q_r \leq \beta$,

$$P_B = \sum_r q_r g_r \leq \beta \sum_r g_r = \beta(y_m - y_1).$$

Apply Proposition 5.1. $\square$

6 Pressure-preserving block stability

Put

$$\delta := \frac{39}{10000}, \quad \beta := \frac{1}{500}, \quad A_m := \delta(m - 6), \quad Q_m := \beta(m - 6).$$

For a block $B = (y_1, \dots, y_m)$, let P_B be the exact pressure (36). Proposition 5.1 gives

$$E_m + P_B \geq A_m, \quad \sum_{r=1}^{m-1} q_r = Q_m. \tag{40}$$

Remark 6.1 (Immediate accounting gain). *Take $m = 262$. Then*

$$A_{262} = \frac{624}{625} < 1, \quad Q_{262} = \frac{64}{125}.$$

The block-stability argument applies directly with P_B : if $P_B \geq A_{262}$ the claim is trivial, while if $P_B < A_{262}$, the positive lower bound $q_r \geq \min_j p_j$ confines the block span to a fixed compact interval and Lemma 4.2 together with (40) gives

$$\mathcal{D}(G_B) + P_B \geq A_{262} - o(1).$$

Averaging the exact pressure as in Section 7 then yields

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq \frac{655000H_{\text{MT}} - 1280}{652504} = 0.6731115225500757511923586 \dots$$

Thus pressure preservation alone yields the intermediate bound displayed above.

6.1 Adjacent-pair pinching

Let $K = (k(y_i - y_j))_{1 \leq i, j \leq m}$ be a limiting Gram matrix. Its positive semidefiniteness follows directly from the integral representation: for every $c \in \mathbb{C}^m$,

$$\sum_{i, j=1}^m \bar{c}_i c_j k(y_i - y_j) = \frac{1}{\sqrt{2} \sin(1/\sqrt{2})} \int_{-1/2}^{1/2} \cos(\sqrt{2}u) \left| \sum_{j=1}^m c_j e^{-2\pi i y_j u} \right|^2 du \geq 0,$$

because $\cos(\sqrt{2}u) > 0$ on $[-1/2, 1/2]$. Hence $K \succeq 0$, $K_{ii} = 1$, and

$$E_m = \text{tr}(K - I)^2 = 2 \sum_{1 \leq i < j \leq m} w(y_j - y_i). \quad (41)$$

Lemma 6.2 (Adjacent-pair defect). *For consecutive gaps $g_j = y_{j+1} - y_j$,*

$$\boxed{\mathcal{D}(K) \geq \sum_{j=1}^{m-1} w(g_j)}. \quad (42)$$

Proof. Apply two separate pinchings to the original matrix K . The first uses the principal 2×2 blocks

$$(1, 2), (3, 4), \dots,$$

while the second, independently applied to the same original K , uses

$$(2, 3), (4, 5), \dots$$

For each of these two pinchings, the spectral trace functional $X \mapsto \text{tr} \Psi(X)$ cannot increase by the standard pinching/majorization principle [8]: pinching is a random-unitary average and this trace functional is convex and unitarily invariant.

The block associated with a consecutive gap g is

$$\begin{pmatrix} 1 & k(g) \\ \overline{k(g)} & 1 \end{pmatrix},$$

whose eigenvalues are $1 \pm |k(g)| \in [0, 2]$. Its defect is therefore exactly $2|k(g)|^2 = 2w(g)$. The first pinching gives

$$\mathcal{D}(K) \geq 2 \sum_{j \text{ odd}} w(g_j),$$

while the second, as a separate inequality for the same K , gives the corresponding bound over even j . Averaging these two independently obtained inequalities proves (42). $\square$

6.2 A scalar kernel inequality

Set

$$g_0 := \frac{89}{100}, \quad t := \frac{33}{2}. \quad (43)$$

Lemma 6.3 (Scalar pressure bound). *For every $g \geq 0$ and every $q \in [0, \beta]$,*

$$w(g) + tqg \geq tqg_0. \quad (44)$$

Proof. The normalized kernel has the integral representation

$$k(g) = \int_{-1/2}^{1/2} \frac{\cos(\sqrt{2}u)}{\sqrt{2} \sin(1/\sqrt{2})} \cos(2\pi gu) du.$$

For $0 \leq x \leq y \leq g_0$ and $|u| \leq 1/2$, one has

$$0 \leq 2\pi x|u| \leq 2\pi y|u| \leq \pi g_0 < \pi.$$

Since cosine is decreasing on $[0, \pi]$ and $\cos(\sqrt{2}u) > 0$ on $[-1/2, 1/2]$, direct comparison of the two integrands gives

$$k(y) \leq k(x).$$

This avoids any differentiation under the integral sign.

The companion Lean development proves the signed bound

$$\frac{171389}{1000000} < k\left(\frac{89}{100}\right). \quad (45)$$

This proof is analytic and does not use an interval enclosure. Writing

$$A = \frac{1}{\sqrt{2}}, \quad B = \frac{89\pi}{100}, \quad \theta = \frac{11\pi}{100},$$

Lean first derives the exact identity

$$K(0.89) = \frac{A \sin A \cos \theta + B \cos A \sin \theta}{B^2 - A^2},$$

then obtains directed rational bounds for the elementary transcendental terms from alternating Taylor estimates and certified bounds for π , proves the denominator positive, and closes the resulting strict rational inequality. As an independent non-Lean reproducibility check, `verify_improved_bound.py` certifies the same signed inequality by exact rational comparisons and rigorous enclosures. In particular $k(g_0) > 0$, and the preceding monotonicity shows that $k(g) \geq k(g_0) > 0$ for $0 \leq g \leq g_0$. Squaring therefore preserves the order on this interval. Moreover exact rational arithmetic gives

$$w\left(\frac{89}{100}\right) > \left(\frac{171389}{1000000}\right)^2 > \frac{2937}{100000} = t\beta g_0. \quad (46)$$

If $0 \leq g \leq g_0$, then

$$w(g) \geq w(g_0) > t\beta g_0 \geq tqg_0.$$

If $g \geq g_0$, then $tqg \geq tqg_0$. This proves (44). □

Lemma 6.4 (Spectral threshold). *Let $K \succeq 0$ be $m \times m$ with diagonal 1, and set*

$$E := \text{tr}(K - I)^2.$$

If $\mathcal{D}(K) < E$, then

$$\mathcal{D}(K) > \frac{m}{m-1}. \quad (47)$$

Proof. If all eigenvalues of K belonged to $[0, 2]$, then $\Psi(\lambda) = (\lambda - 1)^2$ for every eigenvalue and $\mathcal{D}(K) = E$. Hence $\mathcal{D}(K) < E$ implies the existence of an eigenvalue $1 + a > 2$, with $a > 1$.

The remaining $m - 1$ eigenvalues have sum $m - 1 - a$, hence average $1 - a/(m - 1)$. Their nonnegativity implies $a \leq m - 1$, so this average lies in $[0, 1)$. Since Ψ is convex,

$$\mathcal{D}(K) \geq 2a - 1 + (m - 1)\Psi\left(1 - \frac{a}{m - 1}\right) = 2a - 1 + \frac{a^2}{m - 1}.$$

At $a = 1$ the last expression equals $m/(m - 1)$, and it is strictly increasing for $a > 1$, proving (47). $\square$

Proposition 6.5 (Strong block stability at $m = 450$). *Let B be a block of*

$$m = 450$$

consecutive retained simple zeros, with exact pressure P_B from (36). Let G_B denote the corresponding principal Gram block of M° . Put

$$A := A_{450} = \frac{4329}{2500}, \quad Q := Q_{450} = \frac{111}{125}. \quad (48)$$

Uniformly over such blocks,

$$\boxed{\mathcal{D}(G_B) + P_B \geq A - o(1)}. \quad (49)$$

Proof. If $P_B \geq A$, the assertion follows from $\mathcal{D}(G_B) \geq 0$. Assume $P_B < A$. Since $q_r \geq \min_j p_j = 2714/10^7$,

$$\text{span}(B) = \sum_r g_r \leq \frac{P_B}{\min_j p_j} < \frac{A}{\min_j p_j}.$$

Thus all pair separations remain in the fixed compact interval

$$0 \leq |y_i - y_j| \leq R := \frac{A}{\min_j p_j}.$$

Make the uniform error explicit by setting

$$\eta_T(R) := \sup_{\substack{\rho, \rho' \in \mathcal{S}^\circ \\ |x_\rho - x_{\rho'}| \leq R}} |\langle v_\rho, v_{\rho'} \rangle - k(x_\rho - x_{\rho'})|.$$

Lemma 4.1 gives $\eta_T(R) \rightarrow 0$. Since m is fixed and

$$K_B = (k(y_i - y_j))_{i,j=1}^m,$$

the entrywise bound implies

$$\|G_B - K_B\|_{\text{op}} \leq m\eta_T(R).$$

The function Ψ is globally 2-Lipschitz on $[0, \infty)$; therefore Weyl's eigenvalue perturbation inequality gives the quantitative uniform estimate

$$|\mathcal{D}(G_B) - \mathcal{D}(K_B)| \leq 2m\|G_B - K_B\|_{\text{op}} \leq 2m^2\eta_T(R) = o(1). \quad (50)$$

In particular, the $o(1)$ is uniform over every block in the branch $P_B < A$. It remains to prove

$$\mathcal{D}(K_B) + P_B \geq A.$$

Write $K = K_B$ and $D = \mathcal{D}(K)$. By Proposition 5.1 and (41),

$$E + P_B \geq A, \quad E = \text{tr}(K - I)^2.$$

Lemma 6.2 and Lemma 6.3 give

$$D + tP_B \geq \sum_{j=1}^{m-1} (w(g_j) + tq_j g_j) \geq tg_0 \sum_{j=1}^{m-1} q_j = tg_0 Q. \quad (51)$$

Suppose for contradiction that $D + P_B < A$. Since $E + P_B \geq A$, we have $D < E$; Lemma 6.4 therefore yields

$$D > \frac{m}{m-1}.$$

Consequently

$$\begin{aligned} D + tP_B &= t(D + P_B) - (t-1)D \\ &< tA - (t-1)\frac{m}{m-1}. \end{aligned} \quad (52)$$

For $m = 450$, the lower bound (51) contradicts (52), because the exact rational margin is

$$g_0 Q + \left(1 - \frac{1}{t}\right) \frac{450}{449} - A = \frac{6363}{30868750} > 0. \quad (53)$$

Hence $D + P_B \geq A$, and the uniform finite- T statement (49) follows. $\square$

7 Shifted block pinching with exact pressure

Let

$$x_1 < \cdots < x_{S^\circ}$$

be the normalized ordinates of the retained central set S° . As before,

$$S^\circ = N_0^s(T, 2T) - o(N(T, 2T)).$$

For each residue class r of the starting index modulo $m = 450$, first restrict M° to the principal tail beginning at r . The defect of this tail is at most $\mathcal{D}(M^\circ)$, by the same pinching/majorization principle used above. Partition the tail into disjoint full consecutive blocks of size m and one trailing remainder of fewer than m points. Pinching the tail into those principal blocks and the remainder shows that the sum of the full-block defects is at most the tail defect, hence at most $\mathcal{D}(M^\circ)$. As r varies through the m residue classes, every consecutive m -point block with start

index $1, \dots, S^\circ - m + 1$ occurs exactly once. Summing the m pinching inequalities and applying Proposition 6.5 therefore gives

$$m\mathcal{D}(M^\circ) \geq (S^\circ - m + 1)A - \sum_B P_B - o(N). \quad (54)$$

The error here is genuinely $o(N)$ rather than merely blockwise. Since $S^\circ \leq N(T, 2T)$, there are $O(N)$ full blocks in total, and (50) bounds the error of every relevant block by $2m^2\eta_T(R)$ with the same $\eta_T(R) \rightarrow 0$. Hence the accumulated contribution is $O(N\eta_T(R)) = o(N)$. The remainders have nonnegative defect and are simply discarded; the replacement of $S^\circ - m + 1$ by S° later costs only $O(1)$ because m is fixed.

The pressure term retains its exact positional mass. For each fixed relative position r , a given global gap can occupy that position in at most one sliding m -point block. Therefore its coefficient in the sum over all sliding blocks is at most

$$\sum_{r=1}^{m-1} q_r = Q.$$

The span estimate can be made explicit. Because the retained coordinates lie in the sampling interval,

$$x_{S^\circ} - x_1 \leq \frac{LT}{2\pi}.$$

On the other hand Riemann–von Mangoldt gives

$$N(T, 2T) = \frac{T}{2\pi}(L + 2\log 2 - 1) + O(\log T),$$

and $2\log 2 - 1 > 0$. Thus, for all sufficiently large T , $x_{S^\circ} - x_1 \leq N(T, 2T)$, and therefore

$$\sum_B P_B \leq Q(x_{S^\circ} - x_1) \leq QN(T, 2T). \quad (55)$$

Combining (54) and (55), and absorbing $O(1)$ endpoint terms,

$$\mathcal{D}(M^\circ) \geq \frac{A}{m}N_0^s(T, 2T) - \frac{Q}{m}N(T, 2T) - o(N(T, 2T)). \quad (56)$$

For $m = 450$,

$$\mathcal{D}(M^\circ) \geq \frac{481}{125000}N_0^s(T, 2T) - \frac{37}{18750}N(T, 2T) - o(N(T, 2T)). \quad (57)$$

Proof of Theorem 1.1. Write

$$S = N_0^s(T, 2T), \quad N = N(T, 2T).$$

Insert (57) into Corollary 3.3:

$$S \geq H_{\text{MT}}N + \frac{481}{125000}S - \frac{37}{18750}N - o(N).$$

Thus

$$\frac{124519}{125000}S \geq \left(H_{\text{MT}} - \frac{37}{18750}\right)N - o(N).$$

Riemann–von Mangoldt gives $N(T, 2T) \rightarrow \infty$, so $N > 0$ eventually and every $o(N)$ term becomes $o(1)$ after division by N . Dividing by N and passing to the lower limit gives

$$\begin{aligned} \liminf_{T \rightarrow \infty} \frac{S}{N} &\geq \frac{450H_{\text{MT}} - 111/125}{450 - 4329/2500} \\ &= \frac{1125000H_{\text{MT}} - 2220}{1120671} \\ &= 0.6731175265883904388095857434106058666\dots \end{aligned}$$

□

8 Comparison and structural interpretation

At the baseline level, Alpöge–Furman [5] and Lamzouri [7] now provide different unconditional routes to the 67.25%-level Montgomery–Taylor conclusion. The present argument is not obtained by invoking either result: it reconstructs the analytic package used below and then adds the separate stability and local-pressure machinery whose provenance is summarized in the table in Section 1.

Jo’s upstream reproducible draft [19] uses $p_j = 1/3000$ and certifies $19/5000 = 0.0038$, leading to $0.6730085279277797\dots$. Rademacher’s subsequent reproducible refinement [20] keeps the same uniform pressure and strengthens the certified local value to $191/50000 = 0.00382$, leading to $0.6730213619501665\dots$. The position-weighted vector (2) has the same total pressure $1/500$, while Proposition 5.2 proves the stronger local value $39/10000 = 0.0039$.

For orientation, replacing the exact pressure by the coarse span bound

$$P_B \leq \beta \operatorname{span}(B)$$

would give

$$0.6730732086087052768\dots$$

Retaining the exact pressure coefficients q_r until the final shifted-block average gives, without changing the local certificate,

$$0.6731115225500757512\dots$$

The adjacent-pair refinement then gives

$$\boxed{0.6731175265883904388\dots}.$$

The distinction between the two improvements is structural. Pressure preservation removes an avoidable global accounting loss:

$$\beta(m-1) \longrightarrow \beta(m-6).$$

Adjacent-pair pinching supplies new information about the same block Gram matrix without changing the local seven-point certificate. It is precisely this extra defect lower bound that allows the argument to pass beyond the $A_m < 1$ regime and use $m = 450$.

The natural finite-dimensional local optimization remains

$$\sup_{\substack{p_j \geq 0 \\ \sum p_j = 1/500}} \inf_{g \in \mathbb{R}_{\geq 0}^6} \left(\sum_{j=1}^6 p_j g_j + \mathcal{E}_7(g) \right). \quad (58)$$

We do not claim that (2) solves this minimax problem, nor that $m = 450$ is optimal for all refinements of the present method. The theorem uses the already certified vector and a new analytic exploitation of the pressure it carries.

9 Scope and trust base

Logical status of the theorem and relation to the Alpöge–Furman/Claude result. The common compression is fixed in Section 2, and the analytic/Gabor package used in the main argument is proved in Appendices A–D. In particular it is not imported from the Montgomery–Taylor clause of Theorem A of Alpöge–Furman [5]. That clause belongs to the unconditional finite-compression/rank–trace framework discovered by Claude and verified and communicated by Alpöge and Furman [5, 6]. Anthropic’s companion Lean artifact formalizes the corresponding optimized result [9]. Its role here is comparison, provenance, and normalization cross-checking only; no theorem from the Alpöge–Furman paper is a premise of Theorem 1.1.

The analytic external trust base consists of standard classical results: the Weil explicit formula for ζ in the Fourier normalization stated in Appendix C, Stirling’s formula, the Prime Number Theorem, the Riemann–von Mangoldt formula and its standard local zero-count consequence, standard Chebyshev and prime-power estimates, the weighted Montgomery–Vaughan Hilbert inequality, Poisson summation, and standard finite-dimensional matrix facts such as von Neumann’s trace inequality and the pinching/majorization principle [8]. The Montgomery–Vaughan paper is correctly cited as *Hilbert’s inequality*, J. London Math. Soc. (2) **8** (1974), 73–82. The recent pair-correlation papers cited in the introduction provide context; the particular first/second-moment route used here is reconstructed directly in Appendix C.

Proposition 5.2 has two independent certification routes. The primary theorem path is the companion Lean development: the analytic signed-kernel theorem and the exact rational E3 closure combine to prove the seven-point inequality with no interval-computation hypothesis. The finite certificate has the structure

$$511 = 280 + 231, \quad 231 = 208 + 23, \quad 23 = 18 + 5,$$

followed by exact closure of the five remaining words. The independent Arb/FLINT verifier supplies a separate interval-arithmetic check of the same seven-point inequality. Its 128- and 256-bit runs use the same algorithm at different working precisions and therefore count as replications rather than algorithmically independent verifiers.

The formal endpoint is the same quantity as in Theorem 1.1. The Lean development defines

$$R(T) = \frac{(\text{Zeta23.N0simple } T \ (2T) : \mathbb{R})}{(\text{Zeta23.Ncount } T \ (2T) : \mathbb{R})}$$

and proves

$$C_{\text{pub}} \leq \liminf_{T \rightarrow \infty} R(T), \quad C_{\text{pub}} = \frac{1125000H_{\text{MT}} - 2220}{1120671}.$$

Here $\text{Zeta23.Ncount}(T_1, T_2)$ counts nontrivial zeta zeros with $T_1 < \Im \rho \leq T_2$, with analytic multiplicity, while $\text{Zeta23.N0simple}(T_1, T_2)$ counts the simple zeros in the same half-open ordinate window that lie on $\Re \rho = 1/2$. Thus at $(T_1, T_2) = (T, 2T)$ these are exactly the counting functions $N(T, 2T)$ and $N_0^s(T, 2T)$ used in the paper, including the endpoint convention. Eventual positivity of the denominator is derived directly from the formalized Riemann–von Mangoldt theorem, and the trivial counting chain gives $R(T) \leq 1$ eventually.

The transitive axiom audit of the signed-kernel theorem, the seven-point theorem, the epsilon-form theorem, and the literal liminf theorem reports exactly

$$\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}.$$

The CI enforces this exact three-element whitelist for the literal liminf theorem and also rejects `sorryAx`; any additional axiom causes the formal-verification workflow to fail. The formalization is built as an overlay on the pinned Anthropic `formal-math/zeta23` source tree at commit

fbdc36bbf17d20af3fd0447c6d1a8a02773c9844

with the pinned Lean/Mathlib toolchain recorded in the repository. The theorem state used for the formal audit was reproduced by GitHub Actions at project commit

e0f67e112f59b31a4278e1d56377f5e4f1628836

This software provenance is separate from the logical claim that the prose derivation does not invoke the Alpöge–Furman theorem as a premise.

The analytic derivation in the manuscript is self-contained relative to the classical analytic and matrix results listed above. The finite six-gap closure is supplied by the companion Lean certificate rather than reproduced cell-by-cell in prose. Thus the term *self-contained* refers here to the analytic argument and not to a line-by-line transcription of the generated finite certificate.

The pressure vector (2) is an explicit certified improvement, not an optimality theorem. The paper does not claim to reprove the cited classical theorems from first principles, and the formal verification should be read with the software provenance in the preceding paragraph.

A Self-contained derivation of Input I: zero-side inertia

This appendix proves the algebraic finite-compression statement used in Proposition 3.1. The horizontal partner $\rho^\sharp = 1 - \bar{\rho}$ is the composition of conjugation with the functional-equation symmetry; it preserves ordinary ordinate.

A.1 Scope

Let I' be a finite ordinate interval. Since $\rho \mapsto 1 - \bar{\rho}$ preserves ordinary ordinate, the zeros with ordinates in I' form a pairing-invariant finite multiset $Z(I')$ of nontrivial zeta zeros whose ordinates lie in I' . We partition the *distinct* zeros into

$$\mathcal{S}_1 = \{\text{simple critical-line points}\}, \quad \mathcal{S}_2 = \{\text{multiple critical-line points}\},$$

and a set $\mathcal{P}$ containing one representative from each off-line pair

$$\{\rho, \rho^\sharp\}, \quad \rho^\sharp := 1 - \bar{\rho}.$$

Write

$$s_1 = \#\mathcal{S}_1, \quad s_2 = \#\mathcal{S}_2, \quad p = \#\mathcal{P}.$$

The only zeta-specific fact used in the inertia calculation is the combined functional-equation and conjugation symmetry, which preserves multiplicity:

$$m_{\rho^\sharp} = m_\rho.$$

The finite zero-side Hermitian form obtained by compressing Weil’s form to a finite test family can then be written as a pull-back of the block form described below. The present note verifies the resulting inertia statement from scratch.

A.2 Positive index under pull-back

For a Hermitian form H on a finite-dimensional complex vector space, let $n_+(H)$ denote its positive index: the maximal dimension of a subspace on which H is positive definite.

Lemma A.1 (Pull-back monotonicity). *Let H be a Hermitian form on $\mathbb{C}^m$, and let $A : U \rightarrow \mathbb{C}^m$ be either complex-linear or conjugate-linear. Then*

$$n_+(H \circ A) \leq n_+(H).$$

Proof. Let $U_0 \subset U$ be a complex subspace on which $H \circ A$ is positive definite. Then $A|_{U_0}$ is injective. If A is conjugate-linear, $A(U_0)$ is still a complex subspace, because $\alpha A(u) = A(\bar{\alpha} u)$ for every $\alpha \in \mathbb{C}$. Moreover H is positive definite on $A(U_0)$. Hence

$$\dim_{\mathbb{C}} U_0 = \dim_{\mathbb{C}} A(U_0) \leq n_+(H).$$

Taking the maximum over U_0 proves the claim. □

Corollary A.2 (Subadditivity). *For Hermitian forms H_1, H_2 on the same space,*

$$n_+(H_1 + H_2) \leq n_+(H_1) + n_+(H_2).$$

Proof. Apply Lemma A.1 to $H_1 \oplus H_2$ under the diagonal map $u \mapsto (u, u)$. □

A.3 The off-line hyperbolic block

Lemma A.3 (Signature of one horizontal symmetry pair). *Let $m > 0$. On $\mathbb{C}^2$ define*

$$H_m(x, y) = 2m \operatorname{Re}(x\bar{y}).$$

Then

$$H_m(x, y) = \begin{pmatrix} \bar{x} & \bar{y} \end{pmatrix} \begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix},$$

and the form has signature $(1, 1)$.

Proof. The Hermitian matrix

$$B_m = \begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}$$

has characteristic polynomial $\lambda^2 - m^2$, hence eigenvalues m and $-m$. Equivalently, with

$$u = \frac{x+y}{\sqrt{2}}, \quad v = \frac{x-y}{\sqrt{2}},$$

one has

$$H_m(x, y) = m|u|^2 - m|v|^2.$$

Thus $n_+(H_m) = n_-(H_m) = 1$. □

This is the entire source of the “one positive direction per off-line pair” rule. No limiting argument and no assumption concerning the distance of the pair from the critical line is involved.

A.4 The zero-side block form

For each distinct critical-line zero ρ , let m_ρ be its multiplicity. Introduce one evaluation coordinate $z_\rho \in \mathbb{C}$. For each off-line representative $\rho \in \mathcal{P}$, introduce two coordinates $(z_\rho, z_{\rho^\#})$.

Define the Hermitian form

$$\begin{aligned} \mathcal{Q}(z) = & \sum_{\rho \in \mathcal{S}_1 \cup \mathcal{S}_2} m_\rho |z_\rho|^2 \\ & + \sum_{\rho \in \mathcal{P}} 2m_\rho \operatorname{Re}(z_\rho \overline{z_{\rho^\#}}). \end{aligned} \quad (59)$$

Proposition A.4 (Exact block inertia). *The form $\mathcal{Q}$ is an orthogonal direct sum of $s_1 + s_2$ positive one-dimensional blocks and p hyperbolic two-dimensional blocks. Consequently*

$$n_+(\mathcal{Q}) = s_1 + s_2 + p.$$

If the simple critical-line blocks are removed, the remaining form $\mathcal{Q}'$ satisfies

$$n_+(\mathcal{Q}') = s_2 + p.$$

Proof. Every critical-line block is $m_\rho |z|^2$ with $m_\rho > 0$, and hence has positive index one. Every off-line pair block has signature $(1, 1)$ by Lemma A.3. Positive index is additive under orthogonal direct sum, proving both assertions. $\square$

A.5 Compression by evaluation

Let $U = \mathbb{C}^d$ be the coefficient space of the finite family of test functions used to form the compression. For each zero parameter appearing on the zero side, let ℓ_ρ denote the conjugate-linear coefficient functional $c \mapsto \Lambda_\rho(c)$ from Section 2. Define

$$\mathfrak{E}(c) := (\ell_\rho(c))_\rho.$$

The map $\mathfrak{E}$ is conjugate-linear in the displayed coefficient convention; Lemma A.1 was formulated to cover this case directly.

By Proposition 2.1, the normalized finite zero-side matrix $\hat{A}$ is exactly the matrix of the pull-back form

$$c \mapsto \mathcal{Q}(\mathfrak{E}c).$$

Thus this inertia calculation applies to the same matrix $\hat{A}$ used in the main argument.

For a simple critical-line zero $\rho \in \mathcal{S}_1$, use the normalized evaluation vector $v_\rho \in \mathbb{C}^d$ defined in Proposition 2.1; thus the normalized evaluation is $c^* v_\rho$. Proposition 2.1 proves for this same family that

$$\|v_\rho\| \leq 1.$$

Set

$$P_1 = \sum_{\rho \in \mathcal{S}_1} v_\rho v_\rho^*, \quad M = V^* V,$$

where V has the v_ρ as columns, and put

$$Q' = \hat{A} - P_1.$$

Theorem A.5 (Self-contained Input I). *With the preceding notation,*

$$P_1 \succeq 0, \quad \text{rank } P_1 \leq s_1, \quad \text{tr } P_1 \leq s_1,$$

and

$$n_+(Q') \leq s_2 + p.$$

In particular, the finite compression cannot acquire more than one positive direction from each multiple critical-line point and each off-line horizontal symmetry pair after the simple critical-line contribution has been separated.

Proof. Each summand $v_\rho v_\rho^*$ is positive semidefinite and rank at most one, so

$$P_1 \succeq 0, \quad \text{rank } P_1 \leq s_1.$$

Moreover

$$\text{tr } P_1 = \sum_{\rho \in \mathcal{S}_1} \|v_\rho\|^2 \leq s_1.$$

After deleting the simple critical-line coordinates from the block form $\mathcal{Q}$, the remaining form is $\mathcal{Q}'$ of Proposition A.4, with

$$n_+(\mathcal{Q}') = s_2 + p.$$

The matrix Q' on coefficient space is the pull-back of this form under the corresponding restricted evaluation map. Lemma A.1 therefore gives

$$n_+(Q') \leq n_+(\mathcal{Q}') = s_2 + p.$$

□

A.6 Why the functional-equation and conjugation pairing produces the block

For completeness we isolate the algebra behind (59). Weil's Hermitian zero-side form pairs the evaluation at ρ with the evaluation at the horizontal partner under functional-equation plus conjugation symmetry

$$\rho^\sharp = 1 - \bar{\rho}.$$

For an off-line pair of equal multiplicity m_ρ , its two contributions combine as

$$m_\rho \left(\ell_\rho(c) \overline{\ell_{\rho^\sharp}(c)} + \ell_{\rho^\sharp}(c) \overline{\ell_\rho(c)} \right) = 2m_\rho \text{Re} \left(\ell_\rho(c) \overline{\ell_{\rho^\sharp}(c)} \right),$$

which is exactly the hyperbolic block of Lemma A.3. On the critical line, $\rho = \rho^\sharp$, so the same expression collapses to a positive square (up to the normalization convention already absorbed into the zero-side form).

Thus the signature claim is a direct algebraic consequence of functional equation symmetry once the zero-side Hermitian form has been written in evaluation coordinates.

A.7 Adversarial checks

Several possible pathologies do not change the result.

- **Near-line pairs.** As $\operatorname{Re} \rho \rightarrow 1/2$, the two evaluation functionals may become nearly dependent, but pull-back can only *decrease* positive index. No uniform linear independence is required.
- **Deep off-line pairs.** The eigenvalues $\pm m_\rho$ of the abstract pair block are independent of the depth $|\operatorname{Re} \rho - 1/2|$. The evaluation map may distort or annihilate directions, but cannot increase n_+ .
- **Coincident ordinates.** Different zero points may have the same ordinate. This creates dependence among evaluation functionals, which again can only lower the pull-back positive index.
- **Multiplicity.** A multiple critical-line point contributes one positive coordinate block $m_\rho |z|^2$, hence one positive direction, regardless of m_ρ . Multiplicity changes the weight but not its positive index.
- **Noninjective compression.** No injectivity of the full evaluation map is assumed; this is precisely why Lemma A.1 is formulated for arbitrary linear maps.

B Self-contained derivation of Input II: exterior-zero tail

This appendix proves the trace-class tail estimate and its effect on the rank–trace functional. The exterior matrix is grouped into Hermitian critical-line squares and off-line pair terms $uv^* + vu^*$, so that $G = A + E$ holds entrywise under the same zero-side convention.

B.1 Setup

Let

$$\ell = \log \frac{T}{2\pi}, \quad L = \ell, \quad X = e^L = \frac{T}{2\pi}, \quad D_0 = T^{1/2},$$

and let

$$I = (T, 2T], \quad I' = (T - D_0, 2T + D_0].$$

Let the critical sampling grid be

$$\tau_k = T + \frac{2\pi k}{L}, \quad 0 \leq k < d, \quad d = \left\lfloor \frac{LT}{2\pi} \right\rfloor.$$

The tapered Montgomery–Taylor window ϕ has support in $[-L/2, L/2]$, satisfies $0 \leq \phi \leq 1$, and for fixed taper width has

$$C_1 := \|\phi''\|_1 = O(1).$$

Put

$$a = \frac{1}{L} \int_{\mathbb{R}} \phi(u)^2 du.$$

For this window, $a \rightarrow K(0) > 0$; hence $a \asymp 1$.

Use the globally fixed test family (4). Thus

$$G_{jk} = W(f_j, f_k)$$

is exactly the common matrix (8). Using the zero-side expansion of the normalized Weil formula, define A_{jk} by restricting that zero sum to zeros whose ordinary ordinates lie in I' , grouping every off-line zero together with its horizontal partner $\rho^\sharp = 1 - \bar{\rho}$. Define E_{jk} by the complementary exterior zero sum. Since the inside and outside sets form a disjoint partition invariant under $\rho \mapsto \rho^\sharp$, one has entrywise

$$\boxed{G = A + E.}$$

Equivalently,

$$\hat{A} = \hat{G} - \hat{E}, \quad \hat{G} = \frac{G}{aL^2}, \quad \hat{A} = \frac{A}{aL^2}, \quad \hat{E} = \frac{E}{aL^2}.$$

B.2 Off-axis Fourier decay

We use the standard Paley–Wiener integration-by-parts bound.

Lemma B.1 (Complex-shift decay). *Let $\phi \in C_c^2([-L/2, L/2])$. For real $r \neq 0$ and $|y| < 1/2$,*

$$|\hat{\phi}(r - iy)| \leq e^{L|y|/2} \|\phi''\|_1 |r - iy|^{-2} \leq X^{1/4} C_1 |r|^{-2}.$$

Proof. Twice integrating by parts in

$$\hat{\phi}(r - iy) = \int_{-L/2}^{L/2} \phi(u) e^{-iru - yu} du$$

and using the compactly supported C^2 taper gives

$$\hat{\phi}(r - iy) = -(r - iy)^{-2} \int \phi''(u) e^{-i(r - iy)u} du.$$

Since $|e^{-yu}| \leq e^{L|y|/2}$ on the support,

$$|\hat{\phi}(r - iy)| \leq e^{L|y|/2} C_1 |r - iy|^{-2}.$$

Now $|r - iy| \geq |r|$ and, because $|y| < 1/2$,

$$e^{L|y|/2} \leq e^{L/4} = X^{1/4}.$$

□

B.3 One external zero

Let $\rho = \beta + i\gamma$ be a distinct zero with $\gamma \notin I'$ and put

$$y = \beta - \frac{1}{2}, \quad |y| < \frac{1}{2}.$$

Define its sampled vector

$$u_\rho = (\hat{\phi}((\gamma - \tau_k) - iy))_{0 \leq k < d}.$$

Let

$$D = \text{dist}(\gamma, I) \geq D_0.$$

Lemma B.2 (Tail vector bound). *Uniformly for every such zero,*

$$\|u_\rho\|_2^2 \ll C_1^2 X^{1/2} L D^{-3}.$$

Proof. Lemma B.1 gives

$$\|u_\rho\|_2^2 \leq C_1^2 X^{1/2} \sum_{0 \leq k < d} |\gamma - \tau_k|^{-4}.$$

The grid spacing is $h = 2\pi/L$. Comparing the monotone sum with an integral,

$$\sum_{0 \leq k < d} |\gamma - \tau_k|^{-4} \ll D^{-4} + \frac{1}{h} \int_D^\infty t^{-4} dt \ll D^{-4} + LD^{-3} \ll LD^{-3},$$

because $D \geq 1$ and $L \geq 2$ for large T . □

B.4 Summing over zeros outside the enlarged interval

We use only the classical estimate

$$N(t+1) - N(t) \leq A_0 \log(t+3) \tag{60}$$

for some absolute A_0 and all $t \geq 0$.

Lemma B.3 (Weighted exterior zero count). *Counting zeros with multiplicity,*

$$\sum_{\gamma \notin I'} m_\rho \operatorname{dist}(\gamma, I)^{-3} \ll \frac{\log T}{D_0^2}.$$

Proof. For the upper tail, group zeros into intervals

$$2T + D_0 + j < \gamma \leq 2T + D_0 + j + 1, \quad j \geq 0.$$

By (60), the contribution is

$$\ll \sum_{j \geq 0} \frac{\log(2T + D_0 + j + 4)}{(D_0 + j)^3}.$$

Splitting at $j = T$ gives

$$\ll \frac{\log T}{D_0^2}.$$

For the lower positive tail $0 < \gamma < T - D_0$, group backwards from $T - D_0$. The local zero-count estimate is at most $O(\log T)$ throughout this range, so

$$\sum_{0 < \gamma < T - D_0} m_\rho \operatorname{dist}(\gamma, I)^{-3} \ll \log T \sum_{j \geq 0} (D_0 + j)^{-3} \ll \frac{\log T}{D_0^2}.$$

For the negative tail, conjugation symmetry reduces the count in $[-u-1, -u]$ to the same $O(\log(u+3))$ bound as on the positive side. Since its distance from I is at least $T + u$,

$$\sum_{\gamma \leq 0} m_\rho \operatorname{dist}(\gamma, I)^{-3} \ll \sum_{j \geq 0} \frac{\log(j+3)}{(T+j)^3} \ll \frac{\log T}{T^2},$$

which is smaller because $D_0 = T^{1/2}$. Adding the three ranges proves the claim. □

B.5 Trace-norm tail bound

The exterior zero-side matrix must be grouped according to the horizontal symmetry involution. If u_ρ is the sampled evaluation column, then

$$E = \sum_{\substack{\rho \text{ exterior} \\ \rho = \rho^\#}} m_\rho u_\rho u_\rho^* + \sum_{\substack{\{\rho, \rho^\#\} \text{ exterior} \\ \rho \neq \rho^\#}} m_\rho (u_\rho u_{\rho^\#}^* + u_{\rho^\#} u_\rho^*).$$

This is Hermitian term-by-term after pairing. Moreover

$$\|uv^* + vu^*\|_1 \leq 2\|u\|_2\|v\|_2 \leq \|u\|_2^2 + \|v\|_2^2.$$

Hence the scalar tail-vector estimate may be summed exactly as before.

Proposition B.4 (Independent tail estimate). *One has*

$$\|E\|_1 \ll C_1^2 X^{1/2} L \frac{\log T}{D_0^2}.$$

Consequently

$$\|\hat{E}\|_1 = \frac{\|E\|_1}{aL^2} \ll \frac{X^{1/2} \log T}{LD_0^2}.$$

At $\lambda = 1$, $L = \ell$, $X = T/(2\pi)$ and $D_0 = T^{1/2}$, hence

$$\boxed{\|\hat{E}\|_1 \ll T^{-1/2}}.$$

Proof. For critical-line exterior zeros use $\|u_\rho u_\rho^*\|_1 = \|u_\rho\|_2^2$. For each off-line pair use the preceding inequality. Both members have the same ordinate and opposite horizontal displacement, so Lemma B.2 applies to each. Summing with multiplicity and using Lemma B.3 gives the stated unnormalised bound. Division by aL^2 , with $a \asymp 1$, gives the normalized estimate. $\square$

B.6 The rank–trace functional under a trace-class perturbation

Define

$$\mathcal{R}(H) = 4 \operatorname{tr} H - \|H\|_{\mathbb{F}}^2$$

for a finite Hermitian matrix H .

Lemma B.5 (Perturbation inequality). *If $A = G - E$ are Hermitian matrices and $\varepsilon = \|E\|_1$, then*

$$\mathcal{R}(A) \geq \mathcal{R}(G) - 4\varepsilon - 2\|G\|_{\mathbb{F}}\varepsilon - \varepsilon^2. \quad (61)$$

Proof. Since $|\operatorname{tr} E| \leq \|E\|_1 = \varepsilon$,

$$4 \operatorname{tr} A \geq 4 \operatorname{tr} G - 4\varepsilon.$$

Also

$$\|A\|_{\mathbb{F}} \leq \|G\|_{\mathbb{F}} + \|E\|_{\mathbb{F}} \leq \|G\|_{\mathbb{F}} + \|E\|_1 = \|G\|_{\mathbb{F}} + \varepsilon.$$

Squaring and combining the two inequalities proves (61). $\square$

Theorem B.6 (Self-contained Input II). *For the optimized bandwidth $\lambda = 1$, the prime-side moment estimate of Appendix C gives $\|\hat{G}\|_{\mathbb{F}} = O(N(T, 2T)^{1/2})$. Then*

$$\boxed{4 \operatorname{tr} \hat{A} - \|\hat{A}\|_{\mathbb{F}}^2 \geq 4 \operatorname{tr} \hat{G} - \|\hat{G}\|_{\mathbb{F}}^2 - o(N(T, 2T)).}$$

Proof. Apply Lemma B.5 to the normalized matrices with

$$\varepsilon = \|\widehat{E}\|_1 \ll T^{-1/2}$$

from Proposition B.4. Since $N(T, 2T) \asymp T \log T$,

$$\varepsilon = o(N), \quad \varepsilon^2 = o(N),$$

and using the Appendix C bound,

$$\|\widehat{G}\|_F \varepsilon = O\left((T \log T)^{1/2} T^{-1/2}\right) = O((\log T)^{1/2}) = o(N).$$

Thus every perturbation term in (61) is $o(N)$, proving the result. $\square$

Remark B.7 (Package closure). *Appendix C establishes the prime-side second moment without using the present tail comparison. Thus using $\|\widehat{G}\|_F = O(N^{1/2})$ here is logically acyclic.*

C Self-contained derivation of Input III: prime-side moments

This appendix fixes the Fourier normalization of Weil's explicit formula [1, 4], derives the spectral density, controls integrated finite-grid end effects, and evaluates the prime diagonal and both off-diagonal frequency types. The weighted Montgomery–Vaughan inequality [3] is used only with an unspecified absolute constant. The route is deliberately written out from these classical ingredients. Its prime-side architecture parallels Section 5 of Alpöge–Furman [5]—continuous second moment, diagonal prime term, quotient-frequency off-diagonal terms controlled by Montgomery–Vaughan, and secondary terms—but no theorem from that paper is invoked as a premise here. Thus "self-contained" in this appendix is a statement about logical derivation from the cited classical inputs, not a claim of independent discovery of the base analytic architecture.

C.1 Classical input and notation

Put

$$\ell = \log \frac{T}{2\pi}, \quad L = \ell, \quad X = e^L = \frac{T}{2\pi}, \quad I = (T, 2T],$$

and define

$$\ell_1 := \ell + 2 \log 2 - 1. \tag{62}$$

Thus the Riemann–von Mangoldt formula reads

$$N(T, 2T) = \frac{T \ell_1}{2\pi} + O(\ell).$$

Let $\phi = \phi_L \in C_c^2(\mathbb{R})$ be real, even, supported exactly on $[-L/2, L/2]$, with a fixed-width endpoint taper and uniform derivative bounds. Define

$$\begin{aligned} \widehat{\phi}(r) &= \int_{\mathbb{R}} \phi(u) e^{-iru} du, & \Phi(r) &= \int_{\mathbb{R}} \phi(u)^2 e^{-iru} du, \\ a &= \frac{1}{L} \int \phi^2, & b &= \frac{1}{L} \int \phi^4, & g &= \phi^2 * \phi^2. \end{aligned}$$

For the optimized family, a and b remain bounded above and below by positive constants.

We use the classical Weil explicit formula in the following fixed normalization. With

$$\widehat{q}(z) = \int_{\mathbb{R}} q(u) e^{izu} du, \quad q(u) = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{q}(t) e^{-itu} dt,$$

the external theorem used here is

$$\begin{aligned} \sum_{\rho} m_{\rho} \widehat{q}(z_{\rho}) &= \widehat{q}(i/2) + \widehat{q}(-i/2) - \sum_{n \geq 1} \frac{\Lambda(n)}{\sqrt{n}} (q(\log n) + q(-\log n)) \\ &\quad + \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{q}(\tau) \left[\Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \log \pi \right] d\tau. \end{aligned} \quad (63)$$

Here $z_{\rho} = -i(\rho - \frac{1}{2})$ and zeros are counted with multiplicity. This is the standard Guinand–Weil explicit formula in the present Fourier normalization; see Weil [1] and the explicit-formula discussion in Iwaniec–Kowalski [2, §5.5]. We apply it to $q = f * \widetilde{g}$, $\widetilde{g}(u) = \overline{g(-u)}$. Since $f, g \in C_c^2([-L/2, L/2])$, one has $q \in C_c^2([-L, L])$. Moreover

$$\widehat{q}(z) = \mathcal{F}_+ f(z) \overline{\mathcal{F}_+ g(\bar{z})},$$

and two integrations by parts in each factor give, uniformly on every fixed strip $|\Im z| \leq \frac{1}{2} + \varepsilon$,

$$|\widehat{q}(z)| \ll_{\varepsilon, L, f, g} (1 + |z|)^{-4}.$$

Thus the standard strip-decay hypotheses of the explicit formula are satisfied. The support condition also implies that only $n \leq X = e^L$ occur in the prime sum. Fourier inversion then gives the following proposition.

Proposition C.1 (Normalized Weil spectral density). *The compressed Weil form has density*

$$\nu_X(\tau) = \mu(\tau) + \Pi_X(\tau) + P_X(\tau),$$

where

$$\mu(\tau) = \frac{1}{2\pi} \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \frac{\log \pi}{2\pi}, \quad (64)$$

$$P_X(\tau) = -\frac{1}{\pi} \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n), \quad (65)$$

$$\Pi_X(\tau) = \frac{1}{\pi} \Re \frac{X^{1/2+i\tau}}{1/2+i\tau}. \quad (66)$$

More precisely,

$$W(f, g) = \int_{\mathbb{R}} \mathcal{F}_+ f(\tau) \overline{\mathcal{F}_+ g(\tau)} \nu_X(\tau) d\tau.$$

Proof. The gamma contribution in the classical explicit formula is the integral against μ . For the prime term,

$$q(y) + q(-y) = \frac{1}{\pi} \int_{\mathbb{R}} \mathcal{F}_+ f(\tau) \overline{\mathcal{F}_+ g(\tau)} \cos(\tau y) d\tau,$$

and $q(\pm \log n) = 0$ for $n > X$. For the pole terms,

$$H(i/2) + H(-i/2) = \int_{\mathbb{R}} q(u) (e^{-u/2} + e^{u/2}) du.$$

Since $\text{supp } q \subset [-L, L]$, this equals

$$\int_{\mathbb{R}} q(u) e^{-|u|/2} du + \int_{-L}^L q(u) e^{|u|/2} du.$$

With $s = 1/2 + i\tau$,

$$\int_{\mathbb{R}} e^{-|u|/2} e^{-i\tau u} du = (1/4 + \tau^2)^{-1},$$

while

$$\int_{-L}^L e^{|u|/2} e^{-i\tau u} du = 2\Re \frac{X^{\bar{s}} - 1}{\bar{s}} = 2\Re \frac{X^s - 1}{s}.$$

After multiplication by the Fourier-inversion factor $1/(2\pi)$, the $-\Re(1/s)/\pi$ term here cancels the contribution $1/[2\pi(1/4 + \tau^2)]$ from $e^{-|u|/2}$, because $\Re(1/s) = 1/[2(1/4 + \tau^2)]$. Thus the pole density reduces to

$$\Pi_X(\tau) = \frac{1}{\pi} \Re \frac{X^s}{s}, \quad s = \frac{1}{2} + i\tau,$$

which is the formula stated above. This proves the spectral identity with all signs and 2π factors fixed. $\square$

Stirling gives, uniformly for $\tau \in I$,

$$\mu(\tau) = \frac{1}{2\pi} \log \frac{\tau}{2\pi} + O(T^{-2}), \quad \mu'(\tau) = O(T^{-1}).$$

Also

$$|P_X(\tau)| \ll \sqrt{X}, \quad |\Pi_X(\tau)| \ll \frac{\sqrt{X}}{1 + |\tau|},$$

and Riemann–von Mangoldt yields

$$\int_T^{2T} \mu(\tau) d\tau = N(T, 2T) + O(\ell).$$

Let

$$h = \frac{2\pi}{L}, \quad d = \left\lfloor \frac{T}{h} \right\rfloor, \quad \tau_k = T + kh \quad (0 \leq k < d).$$

By (9), the same common matrix G satisfies

$$G_{jk} = \int_{\mathbb{R}} \widehat{\phi}(\tau - \tau_j) \widehat{\phi}(\tau - \tau_k) \nu_X(\tau) d\tau,$$

and set

$$\tilde{G} := \frac{G}{L}, \quad \widehat{G} := \frac{\tilde{G}}{aL} = \frac{G}{aL^2}.$$

This distinction between G , $\tilde{G}$ and $\widehat{G}$ is maintained throughout.

C.2 Fourier and prime-power estimates

The fixed-width C^2 taper gives

$$\max\{|\widehat{\phi}(r)|, |\Phi(r)|\} \leq \psi(r) := \min\left\{L, \frac{C_1}{|r|}, \frac{C_2}{r^2}\right\}, \quad (67)$$

with constants independent of L . Consequently

$$\Psi_0 := \int_0^\infty \psi(r) dr \ll \log L, \quad \int_{\mathbb{R}} \psi(r)^2 |r| dr \ll \log L, \quad \int_{\mathbb{R}} \psi(r)^2 dr \ll L. \quad (68)$$

Plancherel and Fourier inversion give

$$\int_{\mathbb{R}} \Phi(x)^2 dx = 2\pi bL, \quad \int_{\mathbb{R}} \Phi(x)^2 e^{ixy} dx = 2\pi g(y). \quad (69)$$

We also use the classical prime-power estimates

$$\sum_{n \leq x} \frac{\Lambda(n)}{\sqrt{n}} \ll \sqrt{x}, \quad \sum_{n \leq x} \Lambda(n)^2 \ll x \log x, \quad (70)$$

and

$$\sum_{n \leq x} \frac{\Lambda(n)^2}{n} = \frac{1}{2} \log^2 x + O(\log x). \quad (71)$$

The asymptotic (71) follows from the Prime Number Theorem by standard partial summation; the upper bounds in (70) require only Chebyshev-strength estimates. We use Titchmarsh [4] as a standard reference.

C.3 First trace moment

Put

$$S_d(\tau) := \sum_{0 \leq k < d} |\widehat{\phi}(\tau - \tau_k)|^2, \quad S_\infty(\tau) := \sum_{k \in \mathbb{Z}} |\widehat{\phi}(\tau - \tau_k)|^2.$$

For the infinite translated critical grid, Poisson summation gives

$$S_\infty(\tau) = aL^2 \quad (\tau \in \mathbb{R}). \quad (72)$$

We require two distinct end-effect estimates. The first controls the lattice points omitted from the finite grid while τ remains in I :

$$R_{\text{in}} := \int_I (S_\infty(\tau) - S_d(\tau)) |\nu_X(\tau)| d\tau.$$

On I one has

$$|\nu_X(\tau)| \ll B, \quad B := \ell + \sqrt{X},$$

by Stirling and the displayed bounds for P_X, Π_X . Using (67), the left omitted tail is bounded by

$$\sum_{j \geq 1} \int_{jh}^\infty \psi(r)^2 dr,$$

and the right omitted tail is identical up to $O(1)$ endpoint indices. Hence, by Tonelli and (68),

$$R_{\text{in}} \ll B \left(\frac{1}{h} \int_0^\infty r \psi(r)^2 dr + \int_0^\infty \psi(r)^2 dr \right) \ll BL \log L. \quad (73)$$

The second estimate, absent from a naive truncation to I , controls the actual finite-grid mass outside I :

$$R_{\text{out}} := \int_{\mathbb{R} \setminus I} S_d(\tau) |\nu_X(\tau)| d\tau.$$

For the prime and pole pieces the global bounds

$$|P_X(\tau)| \ll \sqrt{X}, \quad |\Pi_X(\tau)| \ll 1 + \sqrt{X}$$

give, by the same endpoint-tail summation,

$$\int_{\mathbb{R} \setminus I} S_d(\tau) (|P_X(\tau)| + |\Pi_X(\tau)|) d\tau \ll \sqrt{X} L \log L. \quad (74)$$

For the archimedean term, Stirling gives $|\mu(\tau)| \ll \ell$ on $|\tau| \leq 3T$. Thus the part of $\mathbb{R} \setminus I$ with $|\tau| \leq 3T$ is $O(\ell L \log L)$ by the same argument. If $|\tau| > 3T$, then every $\tau_k \in [T, 2T + O(h)]$ satisfies $|\tau - \tau_k| \asymp |\tau|$, and (67) gives

$$S_d(\tau) \ll d |\tau|^{-4} \ll TL |\tau|^{-4}, \quad |\mu(\tau)| \ll \log(2 + |\tau|).$$

Consequently

$$\int_{|\tau| > 3T} S_d(\tau) |\mu(\tau)| d\tau \ll TL \int_{3T}^{\infty} \frac{\log(2+t)}{t^4} dt \ll \frac{L\ell}{T^2}.$$

Combining these estimates,

$$R_{\text{out}} \ll (\ell + \sqrt{X}) L \log L + \frac{L\ell}{T^2} = o(L^2 N) \quad (75)$$

for $L \asymp \ell$, $X \asymp T$, since $N \asymp T\ell$.

Now the spectral representation gives the exact decomposition

$$\text{tr } G = \int_I S_d(\tau) \nu_X(\tau) d\tau + \int_{\mathbb{R} \setminus I} S_d(\tau) \nu_X(\tau) d\tau.$$

Using (72), (73), and (75),

$$\text{tr } G = aL^2 \int_I \nu_X(\tau) d\tau + o(L^2 N). \quad (76)$$

By Riemann–von Mangoldt and Stirling,

$$\int_I \mu(\tau) d\tau = N + O(\ell).$$

Termwise integration and (70) give

$$\left| \int_I P_X(\tau) d\tau \right| \ll \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n} \log n} \ll \sqrt{X} = o(N),$$

while the explicit formula for Π_X gives $\int_I \Pi_X(\tau) d\tau = o(N)$. Therefore

$$\text{tr } \tilde{G} = aLN(1 + o(1)), \quad \boxed{\text{tr } \hat{G} = N(1 + o(1))}. \quad (77)$$

C.4 Integrated finite-grid end effects for the second moment

Define

$$K(\tau, \tau') = \sum_{0 \leq k < d} \hat{\phi}(\tau - \tau_k) \hat{\phi}(\tau' - \tau_k),$$

$$K_\infty(\tau, \tau') = \sum_{k \in \mathbb{Z}} \hat{\phi}(\tau - \tau_k) \hat{\phi}(\tau' - \tau_k) = L\Phi(\tau - \tau'),$$

and $K_{\text{out}} = K_\infty - K$. Since $\sum_{k \in \mathbb{Z}} |\hat{\phi}(\tau - \tau_k)|^2 = aL^2$, Cauchy–Schwarz gives

$$|K|, |K_\infty| \leq aL^2, \quad |K_{\text{out}}(\tau, \tau')| \leq \sum_{k \notin [0, d)} \psi(\tau - \tau_k) \psi(\tau' - \tau_k). \quad (78)$$

Put

$$B = C(\ell + \sqrt{X}),$$

so $|\nu_X(\tau)| \leq B$ for $|\tau| \leq 4T$ and $B^2 \ll \ell^2 + X$.

Lemma C.2 (Global density bound used in the tails). *For $|\tau| \leq 4T$,*

$$|\nu_X(\tau)| \ll \ell + \sqrt{X} =: B.$$

If $\tau_k \in [T, 2T]$ and $r = |\tau - \tau_k| > 2T$, then

$$|\nu_X(\tau)| \ll B + \log(r/T).$$

Proof. The prime term satisfies $|P_X| \ll \sqrt{X}$ by $\sum_{n \leq X} \Lambda(n)/\sqrt{n} \ll \sqrt{X}$. The pole density satisfies $|\Pi_X(\tau)| \ll \sqrt{X}/(1 + |\tau|) + 1/(1 + \tau^2)$. Stirling gives $\mu(\tau) \ll 1 + \log(2 + |\tau|)$. This proves the first estimate on $|\tau| \leq 4T$. For $r > 2T$ and $\tau_k \in [T, 2T]$, one has $|\tau| \leq r + 2T < 2r$ and hence $\log(2 + |\tau|) \ll \ell + \log(r/T)$; the prime term remains bounded by $O(\sqrt{X})$ and the pole term is smaller. $\square$

Proposition C.3 (Integrated end effects). *For $I = (T, 2T]$ and*

$$M = \iint_{I \times I} \Phi(\tau - \tau')^2 \nu_X(\tau) \nu_X(\tau') d\tau d\tau',$$

one has

$$\text{tr } \tilde{G}^2 = M + O\left(L\ell \log L (\ell^2 + X)\right), \quad \tilde{G} = G/L. \quad (79)$$

Proof. Expanding $\text{tr } G^2$ gives

$$L^2 \text{tr } \tilde{G}^2 = \iint_{\mathbb{R}^2} |K(\tau, \tau')|^2 \nu_X(\tau) \nu_X(\tau') d\tau d\tau'.$$

On $I \times I$, $|K_\infty|^2 = L^2 \Phi^2$. Write the difference from $L^2 M$ as $E_1 + E_2$, where

$$E_1 = \iint_{I \times I} (|K|^2 - |K_\infty|^2) \nu \nu', \quad E_2 = \iint_{(I \times I)^c} |K|^2 \nu \nu'.$$

For E_1 ,

$$||K|^2 - |K_\infty|^2| \leq |K - K_\infty|(|K| + |K_\infty|) = |K_{\text{out}}|(|K| + |K_\infty|) \ll L^2 |K_{\text{out}}|.$$

Hence

$$|E_1| \ll L^2 B^2 \sum_{k \notin [0, d)} \left(\int_I \psi(\tau - \tau_k) d\tau \right)^2.$$

For omitted grid points at distance $\Delta > 0$ from I ,

$$\int_I \psi(\tau - \tau_k) d\tau \leq \int_{\Delta}^{\infty} \psi(r) dr \leq \min \left\{ \Psi_0, \frac{C}{\Delta} \right\}.$$

At the left edge, omitted points are $k = -j$, $j \geq 1$, and have distance jh from I . At the right edge the two exceptional omitted indices $k = d, d+1$ may lie in, or within one mesh step of, the endpoint $2T$ because $d = \lfloor T/h \rfloor$; for them we use the crude bound $2\Psi_0$. Every further right-hand omitted index has distance at least jh , $j \geq 1$. Therefore

$$\begin{aligned} S &:= \sum_{k \notin [0, d)} \left(\int_I \psi(\tau - \tau_k) d\tau \right)^2 \\ &\ll \Psi_0^2 + \sum_{j \geq 1} \min \left\{ \Psi_0, \frac{C}{jh} \right\}^2. \end{aligned}$$

Let $J = \lceil C/(h\Psi_0) \rceil$. Then

$$\sum_{j \leq J} \Psi_0^2 \ll \frac{\Psi_0}{h} + \Psi_0^2,$$

whereas

$$\sum_{j > J} \frac{C^2}{j^2 h^2} \ll \frac{C^2}{h^2 J} \ll \frac{\Psi_0}{h}.$$

Since $h = 2\pi/L$ and $\Psi_0 = O(\log L)$,

$$S = O(L \log L).$$

Thus

$$|E_1| \ll L^3 B^2 \log L. \tag{80}$$

For E_2 , symmetry lets us place the first variable outside I and multiply by two. Since $|K|^2 \ll L^2 |K|$ and $|K| \leq \sum_{0 \leq k < d} \psi(\tau - \tau_k) \psi(\tau' - \tau_k)$,

$$|E_2| \ll L^2 \sum_{0 \leq k < d} A_k C_k,$$

where

$$A_k = \int_{\tau \notin I} \psi(\tau - \tau_k) |\nu_X(\tau)| d\tau, \quad C_k = \int_{\mathbb{R}} \psi(\tau' - \tau_k) |\nu_X(\tau')| d\tau'.$$

We first bound C_k uniformly. On $|\tau' - \tau_k| \leq 2T$ one has $|\tau'| \leq 4T$, hence

$$\int_{|\tau' - \tau_k| \leq 2T} \psi(\tau' - \tau_k) |\nu_X(\tau')| d\tau' \leq 2B\Psi_0.$$

For $r = |\tau' - \tau_k| > 2T$, one has $|\tau'| \leq 2r$ and

$$|\nu_X(\tau')| \ll B + \log(r/T), \quad \psi(r) \ll r^{-2}.$$

Therefore the far part is

$$\ll \int_{2T}^{\infty} \frac{B + \log(r/T)}{r^2} dr \ll \frac{B+1}{T} \ll B,$$

and hence

$$C_k \ll B\Psi_0. \quad (81)$$

Now sum A_k before integrating. Put

$$\sigma(\tau) = \sum_{0 \leq k < d} \psi(\tau - \tau_k), \quad \Delta = \text{dist}(\tau, I).$$

The distances to grid points dominate $\Delta, \Delta + h, \Delta + 2h, \dots$, so monotonicity yields

$$\sigma(\tau) \leq \psi(\Delta) + \frac{1}{h} \int_{\Delta}^{\infty} \psi(r) dr \ll \psi(\Delta) + L \min \left\{ \Psi_0, \frac{C}{\Delta} \right\}. \quad (82)$$

Also, since $d \ll LT$ and $\psi(r) \ll r^{-2}$,

$$\sigma(\tau) \ll LT \Delta^{-2}. \quad (83)$$

For $0 < \Delta \leq 2T$, $|\tau| \leq 4T$, so $|\nu_X| \leq B$. Integrating (82) on the two sides of I gives

$$\begin{aligned} \int_{0 < \Delta \leq 2T} |\nu_X(\tau)| \sigma(\tau) d\tau &\ll B \left[\Psi_0 + L \int_0^{2T} \min \left\{ \Psi_0, \frac{C}{\Delta} \right\} d\Delta \right] \\ &\ll B [\Psi_0 + L + L \log(2T\Psi_0/C)] \\ &\ll BL\ell. \end{aligned}$$

For $\Delta > 2T$, use (83) and $|\nu_X(\tau)| \ll B + \log(\Delta/T)$:

$$\begin{aligned} \int_{\Delta > 2T} |\nu_X(\tau)| \sigma(\tau) d\tau &\ll LT \int_{2T}^{\infty} \frac{B + \log(\Delta/T)}{\Delta^2} d\Delta \\ &\ll LB. \end{aligned}$$

Consequently

$$\sum_k A_k = \int_{\tau \notin I} |\nu_X(\tau)| \sigma(\tau) d\tau \ll BL\ell. \quad (84)$$

Combining (81) and (84),

$$|E_2| \ll L^2(B\Psi_0)(BL\ell) \ll L^3 B^2 \ell \log L. \quad (85)$$

Equations (80) and (85) give

$$L^2 \text{tr } \tilde{G}^2 = L^2 M + O(L^3 B^2 \ell \log L).$$

Divide by L^2 and use $B^2 \ll \ell^2 + X$ to obtain (79). $\square$

At $L = \ell$ and $X \asymp T$, the error in (79) is indeed $o(TL\ell^2)$: its X -part is $O(T\ell^2 \log \ell)$, smaller than the target $T\ell^3$ by the factor $O(\log \ell / \ell)$, while the remaining $O(\ell^4 \log \ell)$ term is smaller still.

C.5 Evaluation of the continuous second moment

For functions u_1, u_2 on I , write

$$\mathcal{M}[u_1, u_2] = \iint_{I \times I} \Phi(\tau - \tau')^2 u_1(\tau) u_2(\tau') d\tau d\tau'.$$

Then

$$M = \mathcal{M}[\mu, \mu] + \mathcal{M}[P_X, P_X] + 2\mathcal{M}[\mu, P_X] + 2\mathcal{M}[\mu, \Pi_X] + 2\mathcal{M}[P_X, \Pi_X] + \mathcal{M}[\Pi_X, \Pi_X].$$

C.5.1 Archimedean term

Using $\mu(\tau) = \mu(\tau') + O(|\tau - \tau'|/T)$, (68), and (69), one obtains

$$\mathcal{M}[\mu, \mu] = 2\pi bL \int_I \mu(\tau)^2 d\tau + O(\ell^2 \log L) = \frac{bLT\ell_1^2}{2\pi} + o(TL\ell^2). \quad (86)$$

Indeed, the local replacement error is bounded by $O(\ell \int \Phi(x)^2 |x| dx)$ after integrating over I , and the truncation of the full x -integral at the two interval ends costs $O(\ell^2 \int \Phi(x)^2 \min(|x|, T) dx)$.

C.5.2 Prime term: diagonal and two off-diagonal families

Write

$$a_n = \frac{\Lambda(n)}{\sqrt{n}}, \quad y_n = \log n,$$

so

$$P_X(\tau) = -\frac{1}{2\pi} \sum_{n \leq X} a_n (e^{i\tau y_n} + e^{-i\tau y_n}).$$

Substitute $x = \tau - \tau'$. For $|x| < T$, the τ' -interval is $[T + x_-, 2T - x_+]$, $x_{\pm} = \max(\pm x, 0)$. Expanding the four exponential products splits $\mathcal{M}[P_X, P_X]$ into a quotient-frequency diagonal D , a quotient-frequency off-diagonal O_1 , and a product-frequency term O_2 .

For $n = m$, the inner interval has length $T - |x|$, hence Fourier inversion and (68) give

$$D = \frac{T}{\pi} \sum_{n \leq X} a_n^2 g(y_n) + O(L^2 \log L). \quad (87)$$

For product frequencies,

$$\left| \int_{T+x_-}^{2T-x_+} e^{i\tau' \log(nm)} d\tau' \right| \leq \frac{2}{\log(nm)} \leq \frac{2}{\log 4}.$$

Returning to the expression before the x -integration and taking absolute values therefore gives

$$\begin{aligned} |O_2| &\leq \frac{1}{2\pi^2} \sum_{n, m \leq X} a_n a_m \int_{-T}^T \Phi(x)^2 \frac{2}{\log(nm)} dx \\ &\leq \frac{1}{\pi^2 \log 4} \left(\sum_{n \leq X} a_n \right)^2 \int_{\mathbb{R}} \Phi(x)^2 dx. \end{aligned}$$

By the prime-power estimate

$$\sum_{n \leq X} a_n = \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \ll \sqrt{X}$$

and Plancherel

$$\int_{\mathbb{R}} \Phi(x)^2 dx = 2\pi bL \ll L.$$

Consequently

$$\boxed{O_2 \ll XL}. \quad (88)$$

For $n \neq m$, put $\vartheta = y_n - y_m$ and

$$\alpha_n^+ = \int_0^T \Phi(x)^2 e^{ixy_n} dx, \quad \alpha_n^- = \int_{-T}^0 \Phi(x)^2 e^{ixy_n} dx.$$

Then $|\alpha_n^\pm| \leq \pi bL \ll L$. Direct evaluation of the inner τ' -integral gives

$$O_1 = \frac{1}{2\pi^2} \Re \sum_{n \neq m} \frac{a_n a_m}{i(y_n - y_m)} \left[\left(\frac{n}{m} \right)^{2iT} (\alpha_m^+ + \alpha_n^-) - \left(\frac{n}{m} \right)^{iT} (\alpha_n^+ + \alpha_m^-) \right].$$

Thus O_1 is the sum of four Hilbert forms with coefficient pairs

$$(a_n n^{2iT}, a_m m^{-2iT} \alpha_m^+), \quad (a_n n^{2iT} \alpha_n^-, a_m m^{-2iT}), \\ (a_n n^{iT} \alpha_n^+, a_m m^{-iT}), \quad (a_n n^{iT}, a_m m^{-iT} \alpha_m^-).$$

We use the weighted Montgomery–Vaughan Hilbert inequality

$$\left| \sum_{r \neq s} \frac{x_r z_s}{\lambda_r - \lambda_s} \right| \ll \left(\sum_r \frac{|x_r|^2}{\delta_r} \right)^{1/2} \left(\sum_r \frac{|z_r|^2}{\delta_r} \right)^{1/2}, \quad \delta_r = \min_{s \neq r} |\lambda_r - \lambda_s|,$$

with an absolute implied constant; no sharp constant is needed. Only prime powers contribute, because $a_n = \Lambda(n)/\sqrt{n}$. Apply the Montgomery–Vaughan inequality to the distinct frequencies $\lambda_n = \log n$ over those prime powers. If n' is the nearest distinct prime power to n , then $|n' - n| \geq 1$, and $|\log n' - \log n| \geq \log(1 + 1/n) \geq 1/(2n)$ after interchanging the two indices if necessary. Thus $\delta_n^{-1} \leq 2n$. Hence

$$\sum_{n \leq X} \frac{a_n^2}{\delta_n} \ll \sum_{n \leq X} \Lambda(n)^2 \ll XL,$$

and a sequence carrying $\alpha_n^\pm$ has the corresponding weighted square sum $\ll L^3 X$. Each of the four Hilbert forms is therefore $O(L^2 X)$, so

$$O_1 \ll L^2 X. \tag{89}$$

Combining (87)–(89),

$$\mathcal{M}[P_X, P_X] = \frac{T}{\pi} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} g(\log n) + O(L^2 X). \tag{90}$$

By Stieltjes partial summation from (71),

$$\sum_{n \leq X} \frac{\Lambda(n)^2}{n} g(\log n) = \int_0^L y g(y) dy + O(L^2). \tag{91}$$

For completeness, write $A_2(x) = \sum_{n \leq x} \Lambda(n)^2/n = \frac{1}{2} \log^2 x + O(\log x)$. Since the window is uniformly bounded, $\|g\|_\infty \ll L$ and $\|g'\|_\infty \leq \|(\phi^2)'\|_1 \|\phi^2\|_\infty = O(1)$. Integration by parts against the $O(\log x)$ remainder, after the change of variables $y = \log x$, contributes $O(L\|g\|_\infty + \int_0^L y |g'(y)| dy) = O(L^2)$. The main term is exactly $\int_0^L y g(y) dy$.

C.5.3 Mixed and pole terms

Proposition C.4 (Secondary terms). *At $L = \ell$ and $X \asymp T$,*

$$\begin{aligned}\mathcal{M}[\mu, P_X] &\ll \ell\sqrt{X}, & \mathcal{M}[\mu, \Pi_X] &\ll \ell L\sqrt{X}, \\ \mathcal{M}[P_X, \Pi_X] &\ll LX, & \mathcal{M}[\Pi_X, \Pi_X] &\ll \frac{LX}{T}.\end{aligned}$$

All four are $o(TL\ell^2)$.

Proof. Define

$$m(\tau') = \int_I \Phi(\tau - \tau')^2 \mu(\tau) d\tau.$$

Then $\|m\|_\infty \ll \ell L$. Differentiating under the integral and integrating by parts in τ gives

$$\int_I |m'(\tau')| d\tau' \ll \ell L,$$

using $\mu' = O(T^{-1})$ and (69). Therefore, for $y \geq \log 2$,

$$\left| \int_I m(\tau') e^{iy\tau'} d\tau' \right| \ll \frac{\ell L}{y}.$$

Substitution of the defining formula for P_X gives

$$|\mathcal{M}[\mu, P_X]| \ll \ell L \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n} \log n}.$$

The required weighted prime-power bound follows directly from (70): split at $\sqrt{X}$. The range $n \leq \sqrt{X}$ contributes $O(X^{1/4})$, while on $\sqrt{X} < n \leq X$ one has $1/\log n \leq 2/L$, giving $O(\sqrt{X}/L)$. Hence

$$\sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n} \log n} \ll \frac{\sqrt{X}}{L},$$

and therefore $\mathcal{M}[\mu, P_X] \ll \ell\sqrt{X}$.

For the remaining terms use on I

$$|\Pi_X| \ll \frac{\sqrt{X}}{T}, \quad 0 < \mu \ll \ell, \quad |P_X| \ll \sqrt{X},$$

and

$$\sup_\tau \int_I \Phi(\tau - \tau')^2 d\tau' \leq 2\pi bL \ll L.$$

Thus

$$|\mathcal{M}[\mu, \Pi_X]| \ll T \cdot \ell \cdot \frac{\sqrt{X}}{T} \cdot L = \ell L\sqrt{X},$$

$$|\mathcal{M}[P_X, \Pi_X]| \ll T \cdot \sqrt{X} \cdot \frac{\sqrt{X}}{T} \cdot L = LX,$$

and

$$|\mathcal{M}[\Pi_X, \Pi_X]| \ll T \cdot \frac{X}{T^2} \cdot L = \frac{LX}{T}.$$

□

C.6 Scale-free limit

Put $f_L(s) := \phi_L(Ls)^2$. For the fixed-width tapered family used in the paper, f_L is uniformly bounded, even, supported on $[-1/2, 1/2]$, and

$$f_L \rightarrow v$$

in $L^1([-1/2, 1/2])$, where v is bounded, even, and supported on the same interval. The uniform L^∞ bound upgrades this to the convergence needed for the quadratic quantities:

$$a \rightarrow A := \int_{-1/2}^{1/2} v(s) ds, \quad b \rightarrow B_v := \int_{-1/2}^{1/2} v(s)^2 ds.$$

Moreover

$$\frac{g(Lr)}{L} = \int f_L(s) f_L(r-s) ds \rightarrow G_v(r) := \int v(s) v(r-s) ds$$

in $L^1(\mathbb{R})$, because convolution is continuous from $L^1 \times L^1$ to L^1 and the family is uniformly L^1 -bounded. Since all these functions are supported in $[-1, 1]$, multiplication by $|r|$ preserves the convergence. Consequently

$$\begin{aligned} \frac{2}{L^3} \int_0^L y g(y) dy &= 2 \int_0^1 r \frac{g(Lr)}{L} dr \\ &\rightarrow 2 \int_0^1 r G_v(r) dr = J_v, \end{aligned} \tag{92}$$

where

$$J_v := \iint_{[-1/2, 1/2]^2} |s-t| v(s) v(t) ds dt.$$

Combining Proposition C.3, (86), (90), (91), Proposition C.4, and (92),

$$\text{tr } \tilde{G}^2 = \frac{TL}{2\pi} \left(B_v \ell_1^2 + L^2 J_v \right) + o(TL\ell^2).$$

Since, by (62),

$$N(T, 2T) = \frac{T\ell_1}{2\pi} + O(\ell), \quad \frac{L}{\ell_1} \rightarrow 1,$$

and $\hat{G} = \tilde{G}/(aL)$,

$$\|\hat{G}\|_{\text{F}}^2 = \left(\frac{B_v + J_v}{A^2} + o(1) \right) N(T, 2T). \tag{93}$$

Thus

$$c_1(v) = \frac{A^2}{B_v + J_v}.$$

Together with (77), this proves the two required prime-side moments for every profile arising from a uniformly bounded fixed-width tapered family as above. In particular, Appendix D verifies these hypotheses for the Montgomery–Taylor family used in the main argument.

C.7 Closing the Montgomery–Taylor optimization

On $H = L^2([-1/2, 1/2])$ define the self-adjoint integral operator

$$(\mathcal{T}v)(s) = \int_{-1/2}^{1/2} |s - t| v(t) dt.$$

Then

$$c_1(v) = \frac{\langle 1, v \rangle^2}{\langle v, (I + \mathcal{T})v \rangle}.$$

Lemma C.5 (Positivity and invertibility). *$I + \mathcal{T}$ is strictly positive and invertible on H .*

Proof. Schur's test gives

$$\|\mathcal{T}\|_{2 \rightarrow 2} \leq \sup_{|s| \leq 1/2} \int_{-1/2}^{1/2} |s - t| dt = \sup_{|s| \leq 1/2} \left(s^2 + \frac{1}{4} \right) = \frac{1}{2}.$$

Hence

$$\langle v, (I + \mathcal{T})v \rangle \geq (1 - \|\mathcal{T}\|) \|v\|_2^2 \geq \frac{1}{2} \|v\|_2^2.$$

□

Cauchy–Schwarz in the positive inner product $\langle f, g \rangle_A = \langle f, (I + \mathcal{T})g \rangle$ now gives

$$c_1(v) \leq \langle 1, (I + \mathcal{T})^{-1} 1 \rangle,$$

with equality precisely when

$$v \propto (I + \mathcal{T})^{-1} 1.$$

Thus a maximizer satisfies

$$(I + \mathcal{T})v = C \tag{94}$$

for some constant C . The reflection operator $Rv(s) = v(-s)$ commutes with $\mathcal{T}$, and the right-hand side of (94) is even. Since $I + \mathcal{T}$ is invertible, the maximizing solution is therefore even. In the interior,

$$(\mathcal{T}v)'' = 2v,$$

so (94) implies

$$v'' + 2v = 0.$$

Hence

$$v(s) = A_0 \cos(\sqrt{2}s).$$

Conversely, for $v_*(s) = \cos(\sqrt{2}s)$ the function $F = v_* + \mathcal{T}v_*$ satisfies $F'' = 0$ and is even, hence is constant; therefore v_* really satisfies (94), not merely its differentiated equation. Since $1/\sqrt{2} < \pi/2$, $v_* > 0$ on the whole interval, so the constraint $v \geq 0$ is inactive.

Let $\theta = 1/\sqrt{2}$. Then

$$A = \int_{-1/2}^{1/2} v_*(s) ds = \sqrt{2} \sin \theta.$$

Evaluating $C = (I + \mathcal{T})v_*$ at $s = 0$ gives

$$C = 1 + 2 \int_0^{1/2} t \cos(\sqrt{2}t) dt = \cos \theta + \frac{\sin \theta}{\sqrt{2}}.$$

Since $\langle v_*, (I + \mathcal{T})v_* \rangle = CA$,

$$c_1^* = \frac{A}{C} = \frac{2 \tan(1/\sqrt{2})}{\sqrt{2} + \tan(1/\sqrt{2})},$$

and therefore

$$\boxed{\frac{1}{c_1^*} = \frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}.} \quad (95)$$

D Self-contained derivation of Input IV: Poisson–Gabor sampling

This appendix proves the critical-lattice Poisson identity, the tapered Fourier decay, finite-grid truncation, and the Montgomery–Taylor overlap limit. The window is real and even, so the real-axis transforms used below are real and even; modulus squares may be retained throughout.

D.1 Statement to be verified

Let $L \rightarrow \infty$, put

$$h = \frac{2\pi}{L}, \quad d = \left\lfloor \frac{T}{h} \right\rfloor, \quad \tau_k = T + kh.$$

The finite sampling grid is

$$\{\tau_j : 0 \leq j < d\},$$

and the infinite grid is its extension to $k \in \mathbb{Z}$. This is exactly the translated critical lattice used in the main argument. The translation plays no role in the aliasing calculation because all nonzero dual-lattice terms vanish. and let $\phi_L \in C_c^2(\mathbb{R})$ be a real tapered window with exact support $[-L/2, L/2]$. Write

$$\widehat{\phi}_L(\xi) = \int_{\mathbb{R}} \phi_L(u) e^{-i\xi u} du$$

and

$$a_L = \frac{1}{L} \int_{\mathbb{R}} \phi_L(u)^2 du.$$

For completeness, fix once and for all a smooth cutoff $\chi \in C^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$, $\chi(t) = 0$ for $t \leq 0$, $\chi(t) > 0$ for $t > 0$, and $\chi(t) = 1$ for $t \geq 1$. For all sufficiently large L , one concrete Montgomery–Taylor window satisfying the conventions used throughout the paper is

$$\phi_L(u) = \sqrt{\cos\left(\frac{\sqrt{2}u}{L}\right)} \chi\left(\frac{L}{2} - |u|\right).$$

The cosine is strictly positive on $[-L/2, L/2]$ because $1/\sqrt{2} < \pi/2$. The cutoff is identically one away from the unit-width endpoint layers and flat where it meets the support boundary, so $\phi_L \in C_c^2(\mathbb{R})$ is real, even, uniformly bounded, has support exactly $[-L/2, L/2]$, and has uniformly controlled first and second taper derivatives. In particular,

$$\phi_L(u)^2 = \cos\left(\frac{\sqrt{2}u}{L}\right), \quad |u| \leq \frac{L}{2} - 1.$$

Any fixed-width smooth taper with these properties gives the same limiting profile and the same leading constants. Define

$$K(x) = \int_{-1/2}^{1/2} \cos(\sqrt{2}t) \cos(2\pi xt) dt, \quad k(x) = \frac{K(x)}{K(0)}.$$

The required form of Input IV is the following.

Theorem D.1 (Self-contained Input IV). *Fix $R > 0$. Let γ, γ' lie inside the finite sampling interval and have normalized coordinates*

$$x = \frac{\gamma - T}{h} = \frac{L(\gamma - T)}{2\pi}, \quad x' = \frac{\gamma' - T}{h} = \frac{L(\gamma' - T)}{2\pi}.$$

Assume both points are at least L^2 grid cells from each finite-grid endpoint and that $|x - x'| \leq R$. Then the normalized finite-grid Gram entry satisfies

$$\frac{1}{a_L L^2} \sum_{\tau_j \text{ in the finite grid}} \widehat{\phi}_L(\gamma - \tau_j) \overline{\widehat{\phi}_L(\gamma' - \tau_j)} = k(x - x') + o(1),$$

uniformly for $|x - x'| \leq R$.

D.2 The infinite-grid identity

Lemma D.2 (Critical-grid Poisson identity). *Assume $\phi \in C_c^2(\mathbb{R})$ is supported in an interval of length at most L , and let $\tau_k = T + 2\pi k/L$. Then*

$$\sum_{k \in \mathbb{Z}} \widehat{\phi}(\tau - \tau_k) \overline{\widehat{\phi}(\tau' - \tau_k)} = L \int_{\mathbb{R}} |\phi(u)|^2 e^{-i(\tau - \tau')u} du.$$

Proof. Set

$$F_{\tau, \tau'}(\xi) = \widehat{\phi}(\tau - \xi) \overline{\widehat{\phi}(\tau' - \xi)}.$$

With the Fourier convention above, a direct Fourier inversion calculation gives

$$\widehat{F}_{\tau, \tau'}(s) = 2\pi e^{-i\tau' s} \int_{\mathbb{R}} \phi(u) \overline{\phi(u - s)} e^{-i(\tau - \tau')u} du.$$

Because $\phi \in C_c^2$, the function $F_{\tau, \tau'}$ decays like $O(|\xi|^{-4})$, while its Fourier transform is compactly supported; hence the Poisson summation formula applies absolutely. On the shifted lattice $T + h\mathbb{Z}$, $h = 2\pi/L$, it gives

$$\sum_{k \in \mathbb{Z}} F_{\tau, \tau'}(T + kh) = \frac{1}{h} \sum_{m \in \mathbb{Z}} e^{imLT} \widehat{F}_{\tau, \tau'}(mL).$$

Because ϕ is supported in an interval of length at most L , the translated supports of $\phi(u)$ and $\phi(u - mL)$ are disjoint for $m \neq 0$, apart from endpoints of measure zero. Thus every nonzero dual-lattice term vanishes; the phase e^{imLT} is therefore irrelevant. Since $1/h = L/(2\pi)$,

$$\sum_{k \in \mathbb{Z}} F_{\tau, \tau'}(T + kh) = L \int_{\mathbb{R}} |\phi(u)|^2 e^{-i(\tau - \tau')u} du.$$

□

Remark D.3. *The exact support condition $\text{supp } \phi_L = [-L/2, L/2]$ is the one used by the optimized fixed-width tapered family. Hence the critical-density Poisson identity has no aliasing term: shifts by mL , $m \neq 0$, overlap only on sets of measure zero.*

D.3 Fourier decay

Lemma D.4 (Uniform decay). *Suppose $\phi_L \in C_c^2([-L/2, L/2])$ is uniformly bounded and its taper has fixed physical width with uniformly bounded first and second derivatives. Then*

$$|\widehat{\phi}_L(r)| \ll \min \left\{ L, \frac{1}{|r|}, \frac{1}{r^2} \right\},$$

with an implied constant independent of L .

Proof. The L^1 estimate gives $|\widehat{\phi}_L(r)| \ll L$. One integration by parts, using compact support and the bounded total variation of ϕ_L , gives $|\widehat{\phi}_L(r)| \ll |r|^{-1}$. A second integration by parts on each smooth piece gives $|\widehat{\phi}_L(r)| \ll |r|^{-2}$: the bulk derivatives of the Montgomery–Taylor profile contribute $O(L^{-1})$ and $O(L^{-2})$ per unit length, while the fixed-width taper contributes $O(1)$ total variation at each derivative level. All boundary terms vanish because the taper is compactly supported and C^2 . $\square$

D.4 Finite-grid truncation

Lemma D.5 (Pointwise tail). *Fix $R > 0$. Suppose γ, γ' are at least L^2 grid cells from a finite-grid endpoint and $|L(\gamma - \gamma')/(2\pi)| \leq R$. Then the contribution of the omitted half-grid beyond that endpoint is $O(L^{-2})$ before Gram normalization.*

Proof. Let the omitted index be $n \geq L^2$ cells beyond the nearer endpoint. Since $h = 2\pi/L$ and the normalized separation of γ, γ' is bounded,

$$|\gamma - \tau_j| \asymp \frac{n}{L}, \quad |\gamma' - \tau_j| \asymp \frac{n}{L}$$

uniformly in the stated region. Lemma D.4 therefore gives

$$|\widehat{\phi}_L(\gamma - \tau_j)\widehat{\phi}_L(\gamma' - \tau_j)| \ll \frac{L^4}{n^4}.$$

Consequently

$$\sum_{n \geq L^2} \frac{L^4}{n^4} \ll L^4 \int_{L^2-1}^{\infty} t^{-4} dt = O(L^{-2}).$$

The same estimate applies at the other endpoint. $\square$

D.5 Montgomery–Taylor overlap limit

Lemma D.6 (Normalization and overlap). *For the fixed-width tapered Montgomery–Taylor family,*

$$a_L = K(0) + o(1), \quad K(0) = \int_{-1/2}^{1/2} \cos(\sqrt{2}t) dt = \sqrt{2} \sin(1/\sqrt{2}) > 0.$$

Moreover, uniformly for $|x| \leq R$,

$$\frac{1}{a_L L} \int_{\mathbb{R}} \phi_L(u)^2 e^{-i(2\pi x/L)u} du = k(x) + o(1).$$

Proof. Put $u = Lt$ and $f_L(t) = \phi_L(Lt)^2 \mathbf{1}_{[-1/2, 1/2]}(t)$. Outside a set of t -measure $O(L^{-1})$ arising from the fixed-width taper,

$$f_L(t) = \cos(\sqrt{2}t).$$

Uniform boundedness of the window therefore gives

$$\|f_L - \cos(\sqrt{2}\cdot)\|_{L^1([-1/2, 1/2])} = O(L^{-1}).$$

In particular, for every real x ,

$$\left| \int_{-1/2}^{1/2} (f_L(t) - \cos(\sqrt{2}t)) e^{-i2\pi xt} dt \right| \leq O(L^{-1}),$$

so the Fourier convergence is uniform on every compact set (indeed uniformly in all x). Consequently

$$a_L = \int_{-1/2}^{1/2} \cos(\sqrt{2}t) dt + o(1) = K(0) + o(1)$$

and

$$\frac{1}{L} \int \phi_L(u)^2 e^{-i(2\pi x/L)u} du = \int_{-1/2}^{1/2} \cos(\sqrt{2}t) e^{-i2\pi xt} dt + o(1).$$

The odd imaginary part vanishes. Since $K(0) = \sqrt{2} \sin(1/\sqrt{2}) > 0$, one also has $a_L \geq K(0)/2$ for all sufficiently large L . Division by a_L is therefore uniform and proves the claim. $\square$

The overlap has the explicit form

$$K(x) = \frac{\sin(\pi x - 1/\sqrt{2})}{2\pi x - \sqrt{2}} + \frac{\sin(\pi x + 1/\sqrt{2})}{2\pi x + \sqrt{2}},$$

with removable singularities interpreted by continuity.

Proof of Theorem D.1. Apply Lemma D.2 to the infinite grid. By Lemma D.5, replacing the infinite grid by the finite central grid changes the unnormalized sum by $O(L^{-2})$. Since $a_L \asymp 1$, division by $a_L L^2$ makes this error $O(L^{-4})$. The remaining infinite-grid expression, after normalization, is the overlap in Lemma D.6. This gives $k(x - x') + o(1)$ uniformly on every fixed compact set of normalized separations. $\square$

D.6 Audit notes and exact scope

The proof above is independent of the zeta explicit formula, zero density, and prime-side pair correlation. It uses only:

1. Poisson summation on the critical lattice $2\pi\mathbb{Z}/L$;
2. compact support and regularity of the chosen window;
3. the fixed-width taper;
4. elementary integration by parts.

Thus Input IV is proved independently from Poisson summation and elementary Fourier estimates for the exact support/taper convention used in the main manuscript.

E Certificate metadata and formal verification

Formal theorem correspondence. The companion Lean development formalizes the same dyadic ratio and constant as Theorem 1.1. In the formal statement, the ratio is

$$R(T) = \frac{(\text{Zeta23.N0simple } T \ (2T) : \mathbb{R})}{(\text{Zeta23.Ncount } T \ (2T) : \mathbb{R})}$$

and the certified constant is

$$C_{\text{pub}} = \frac{1125000H_{\text{MT}} - 2220}{1120671}.$$

The kernel-checked endpoint has the type

$$C_{\text{pub}} \leq \liminf_{T \rightarrow \infty} R(T).$$

Thus the numerator, denominator, constant, and limiting operation match the objects displayed in Theorem 1.1; this is the statement-alignment bridge between the paper theorem and the formal theorem.

Seven-point statement alignment. The formal local theorem is the same universal statement as Proposition 5.2: for every six-tuple $g = (g_1, \dots, g_6) \in \mathbb{R}_{\geq 0}^6$,

$$\sum_{j=1}^6 p_j g_j + \sum_{r=1}^6 \frac{2}{7-r} \sum_{i=1}^{7-r} w(g_i + \dots + g_{i+r-1}) \geq \frac{39}{10000},$$

with

$$p = 10^{-7}(2714, 3733, 3553, 3553, 3733, 2714)$$

and with $w = k^2$ for the normalized limiting kernel of Section 4. Thus the formal finite certificate, the local inequality used by Proposition 5.1, and the displayed Proposition 5.2 have identical mathematical data.

Finite-certificate map. The analytically unresolved region is exactly

$$g_j > 0.89 \quad (1 \leq j \leq 6), \quad g_1 + \dots + g_6 < 14.37.$$

The one-body localization partitions the six positions according to the symmetric pattern

$$A - B - C - C - B - A.$$

Basin labels are numbered from 1 in the displayed word codes, so a digit such as 2 selects the second entry of the corresponding A , B , or C basin list. There are respectively 7, 5, 6, 6, 5, 7 rational basin choices, hence

$$7 \cdot 5 \cdot 6 \cdot 6 \cdot 5 \cdot 7 = 44100$$

raw basin words. The exact basin intervals are

$$\begin{aligned} A : & [955, 1158]/1000, [1792, 2258]/1000, [2612, 3396]/1000, \\ & [3502, 4465]/1000, [4500, 5466]/1000, [5500, 6413]/1000, \\ & [6576, 7270]/1000, \\ B : & [955, 1158]/1000, [1798, 2249]/1000, [2637, 3358]/1000, \\ & [3568, 4375]/1000, [4611, 5285]/1000, \\ C : & [955, 1158]/1000, [1797, 2251]/1000, [2633, 3365]/1000, \\ & [3556, 4392]/1000, [4580, 5322]/1000, [5793, 6078]/1000. \end{aligned}$$

The one-body exclusion lemmas prove that every strict counterexample in the hard core places each coordinate in the appropriate listed basin; the gaps between these intervals are therefore excluded analytically rather than silently omitted from the enumeration. The certified one-body floors, in units of 10^{-6} , are

$$\begin{aligned}\mu_A &= (286, 550, 817, 1085, 1353, 1622, 1890), \\ \mu_B &= (394, 756, 1123, 1490, 1859), \\ \mu_C &= (375, 720, 1069, 1419, 1770, 2122).\end{aligned}$$

Their meaning is pointwise. If a coordinate g_i lies in basin j of its positional type $X \in \{A, B, C\}$, then the formal micro-floor theorem gives

$$10^{-6}\mu_X(j) \leq p_i g_i + \frac{1}{3}w(g_i).$$

Summing the six inequalities therefore gives a lower bound for the linear pressure plus the six adjacent-pair contributions, hence for the full functional $\mathcal{F}_p$ because every remaining pair term is nonnegative. For a word (a, b, c, d, e, f) its one-body micro-energy is

$$\mu_A(a) + \mu_B(b) + \mu_C(c) + \mu_C(d) + \mu_B(e) + \mu_A(f).$$

A strict counterexample must have this integer sum < 3900 , because $3900 \cdot 10^{-6} = 39/10000$. Exact finite enumeration of this condition reduces the 44100 raw words to precisely 511.

Each subsequent round is a rational quadratic certification step. For every current box and every one of the 21 consecutive partial sums $s_{i,r} = g_i + \dots + g_{i+r-1}$, the analytic cell catalog supplies a kernel-checked lower minorant of the form

$$w(s_{i,r}) \geq \alpha (s_{i,r} - q)^2 - \eta,$$

with rational α, q, η . After multiplying by the coefficient $2/(7-r)$ and adding the exact pressure term, these minorants assemble to a six-variable rational quadratic

$$Q_B(g) \leq \mathcal{F}_p(g)$$

throughout the box B . Exact LDL^t completion writes

$$Q_B(g) = q_B + \sum_j d_j L_j (g - g_B^*)^2, \quad d_j > 0.$$

If $q_B \geq 39/10000$, the whole box is discarded. Otherwise any strict counterexample in B must satisfy

$$\sum_j d_j L_j (g - g_B^*)^2 < \frac{39}{10000} - q_B.$$

Weighted Cauchy–Schwarz then gives rational bounds for each partial sum $s_{i,r}$; outward rational rounding produces a smaller certified box, which is the input to the next round. Thus the *surviving* cases are not sampled points: they are exact nested rational boxes containing every possible strict counterexample left by the previous round.

The three rounds have the following kernel-checked accounting:

round	input boxes	discarded	contracted survivors	distinct minorant cells
R_0	511	280	231	190
R_1	231	208	23	2433
R_2	23	18	5	246

The cell counts refer to distinct one-variable interval certificates; a single certified cell can be reused in many of the 21 block positions and many six-dimensional boxes.

Coverage of each round is itself proved in Lean, rather than trusted to the certificate generator. The discard-code sets satisfy the exact set identities

$$D_0 = S_{511} \setminus S_{231}, \quad D_1 = S_{231} \setminus S_{23}, \quad D_2 = S_{23} \setminus S_5,$$

and the membership routers prove that a strict counterexample in the input set cannot belong to any discard code. It must therefore lie in the next survivor set. In particular, the three cardinality equalities

$$511 = 280 + 231, \quad 231 = 208 + 23, \quad 23 = 18 + 5$$

are not merely generator diagnostics; they are part of the checked finite routing.

The Python scripts are certificate producers, not trusted proof components. They compute rational cells, quadratic coefficients, minimizers and proposed contractions, but the generated Lean files recheck the interval hypotheses, the one-variable minorants, the algebraic quadratic identities, positivity of the LDL^t square coefficients, the discard inequalities, the contraction implications and the membership routing. A faulty generated datum therefore causes elaboration or kernel checking to fail rather than becoming an additional mathematical assumption.

The five terminal words are

$$121212, \quad 121221, \quad 122121, \quad 212121, \quad 212212.$$

For these five boxes the final layer replaces the bootstrap minorants by strong-convexity lower bounds for the limiting weight. The required curvature and anchor inequalities are subdivided into 113 exact rational interval cells, which yield ten reusable strong lower bounds. For each terminal word, 21 such block bounds assemble to a final rational quadratic below $\mathcal{F}_p$. Exact LDL^t completion proves that the minimum of each final quadratic is strictly larger than $39/10000$.

The five exact margins $Q_{\min} - 39/10000$ are

$$\begin{aligned} m_{121212} = m_{212121} &= \frac{1191001492104159283646995579}{1821262686688912135146560000000000} = 6.5394272929921 \dots \times 10^{-7}, \\ m_{121221} = m_{122121} &= \frac{122261493153092324461322292173429}{13654115346619578930629550480000000000} = 8.9541863423148 \dots \times 10^{-7}, \\ m_{212212} &= \frac{10429603649012763691}{1397723775304050000000000} = 7.4618489241510 \dots \times 10^{-6}. \end{aligned}$$

In particular, all five margins exceed $1/2000000$, exactly as checked by the generated Lean inequalities. Consequently no strict counterexample remains, and the composed certificate establishes

$$44100 \longrightarrow 511 \longrightarrow 231 \longrightarrow 23 \longrightarrow 5 \longrightarrow \emptyset.$$

Together with the analytic signed-kernel theorem, this yields Proposition 5.2.

Formal verification snapshot. The formal theorem state used for this manuscript is identified by project commit

$$\text{e0f67e112f59b31a4278e1d56377f5e4f1628836}$$

on top of the pinned `formal-math/zeta23` source tree. GitHub Actions reconstructed the overlay, regenerated the complete E3 stack, kernel-checked the literal `liminf` theorem, and independently typechecked the endpoint wrapper. The reported axiom closure is

$$\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}.$$

The workflow checks equality with this exact whitelist and fails on any additional axiom or on `sorryAx`.

Independent Arb/FLINT check. The Arb/FLINT verifier supplies a separate reproducibility check of Proposition 5.2. The reproducibility artifact [23] records the verifier, execution log, environment, cryptographic hashes, and reproduction instructions. The 256-bit run reports **verified=true**, with 1,119,372 visited nodes, 559,038 splits, and maximum depth 39; the same verifier also succeeds at 128-bit working precision. Because both runs use the same implementation, the two precisions are replications rather than independent algorithms.

Field	256-bit interval check
verified	true
target	$F_{6,\text{nonuniform}} \geq 39/10000$
grid	4000
precision	256 bits
nodes	1,119,372
pruned	560,334
splits	559,038
maximum depth	39
initial boxes	1296
interval prunes	364,112
tangent prunes	190,953
pressure prunes	5,269
elapsed time	201.487482 s

The pressure vector recorded by the verifier is

$$(2714, 3733, 3553, 3553, 3733, 2714)/10^7.$$

F Exact arithmetic

For the main refinement,

$$m = 450, \quad A = \frac{39}{10000}(m - 6) = \frac{4329}{2500}, \quad Q = \frac{1}{500}(m - 6) = \frac{111}{125}.$$

Hence

$$\frac{A}{m} = \frac{481}{125000}, \quad \frac{Q}{m} = \frac{37}{18750}, \quad 1 - \frac{A}{m} = \frac{124519}{125000}.$$

Therefore

$$\frac{H_{\text{MT}} - Q/m}{1 - A/m} = \frac{450H_{\text{MT}} - 111/125}{450 - 4329/2500} = \frac{1125000H_{\text{MT}} - 2220}{1120671}.$$

The scalar adjacent-pair input is

$$g_0 = \frac{89}{100}, \quad t = \frac{33}{2}, \quad \frac{171389}{1000000} < k(g_0),$$

which implies

$$w(g_0) > \left(\frac{171389}{1000000} \right)^2 > \frac{2937}{100000} = t\beta g_0.$$

The contradiction margin is the positive rational

$$g_0Q + \left(1 - \frac{1}{t}\right) \frac{450}{449} - A = \frac{6363}{30868750} > 0.$$

As an independent non-Lean reproducibility check, the accompanying script `verify_improved_bound.py` verifies these identities and rigorously encloses the transcendental quantities entering the final decimal bound; the signed kernel inequality itself is proved analytically in the companion Lean development and is incorporated into the formal theorem path.

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